QUESTION 1 [M] Define a solution and distinguish between a saturated, unsaturated and supersaturated solution.

ANSWER 1

A solution is a homogeneous mixture. Unsaturated solutions can dissolve more solute at the stated temperature; saturated solutions are at equilibrium with undissolved solute; supersaturated solutions contain more dissolved solute than the equilibrium solubility and are metastable.

QUESTION 2 [M] Explain why a solution of a non-volatile solute has a lower vapour pressure than the pure solvent.

ANSWER 2

For a non-volatile solute, only solvent contributes appreciably to vapour. Its mole fraction is below one, so Raoult's law gives $p_{\text{solvent}} = x_{\text{solvent}}\, p^\circ_{\text{solvent}} < p^\circ_{\text{solvent}}$.

QUESTION 3 [M] State Raoult's law for a solution containing two volatile components.

ANSWER 3

For ideal binary volatile liquids A and B: $p_A = x_A p_A^\circ$, $p_B = x_B p_B^\circ$ *and* $P_{\text{total}} = p_A + p_B = x_A p_A^\circ + x_B p_B^\circ$.

QUESTION 4 [H] Derive the relation between relative lowering of vapour pressure and mole fraction of a non-volatile solute.

ANSWER 4

For solvent A and non-volatile solute B, $p_A = x_A p_A^\circ$. *Thus* $(p_A^\circ - p_A)/p_A^\circ = 1 - x_A = x_B$, *the solute mole fraction. This assumes an ideal dilute solution and negligible solute vapour pressure.*

QUESTION 5 [M] What is an ideal solution? Give two conditions for ideal behaviour.

ANSWER 5

An ideal solution obeys Raoult's law at every composition. Mixing has $\Delta H_{\text{mix}} = 0$ *and* $\Delta V_{\text{mix}} = 0$; *unlike and like intermolecular interactions are approximately equal.*

QUESTION 6 [H] Explain positive and negative deviations from Raoult’s law with suitable examples.

ANSWER 6

Positive deviation occurs when A–B attractions are weaker than A–A/B–B, raising vapour pressure (e.g. ethanol–acetone approximately). Negative deviation occurs when A–B attractions are stronger, lowering vapour pressure (e.g. acetone–chloroform).

QUESTION 7 [M] Why do some liquid pairs show positive deviation from Raoult’s law?

ANSWER 7

Unlike molecules attract one another less strongly than like molecules; they escape more readily to vapour, so the observed vapour pressure exceeds Raoult’s-law prediction.

QUESTION 8 [H] A liquid mixture shows negative deviation from Raoult’s law. Predict the signs of ΔHmix and ΔVmix and justify.

ANSWER 8

Negative deviation indicates stronger A–B attractions. Mixing is generally exothermic, $\Delta H_{mix}<0$, and contraction commonly gives $\Delta V_{mix}<0$.

QUESTION 9 [M] Define molarity, molality, mole fraction and mass percentage.

ANSWER 9

Molarity $M=\text{moles solute}/L$ solution; molality $m=\text{moles solute}/kg$ solvent; mole fraction $x_i=n_i/\Sigma n$; mass percent $=100(\text{mass solute}/\text{mass solution})$.

QUESTION 10 [H] Why is molality preferred over molarity when studying colligative properties?

ANSWER 10

Molality is based on solvent mass and is essentially independent of temperature, unlike molarity, whose volume basis changes with temperature.

QUESTION 11 [M] Calculate the molarity of a solution containing 5.85 g NaCl in 500 mL of solution.

ANSWER 11 · NEEDS QUALIFICATION

$n(NaCl) = 5.85/58.44 = 0.1001$ mol; $V = 0.500$ L. $M = 0.1001/0.500 = 0.200$ mol L^{-1}. Because "500 mL" has ambiguous trailing-zero precision, report 0.2, 0.20, or 0.200 M if it is intended to 1, 2, or 3 significant figures, respectively.

QUESTION 12 [H] A solution is prepared by dissolving 18 g glucose in 180 g water. Calculate the mole fractions of glucose and water.

ANSWER 12 · NEEDS QUALIFICATION

$n(\text{glucose}) = 18/180 = 0.100$ mol; $n(H_2O) = 180/18 = 10.0$ mol. $x_{\text{glucose}} = 0.00990099\ldots$ and $x_{\text{water}} = 0.990099\ldots$. If 180 g water is intended to 2 significant figures (matching 18 g glucose), report 0.0099 and 0.99; if 180 g is only 1 significant figure, round both to 1 significant figure.

QUESTION 13 [M] Define solubility and state the effect of temperature on the solubility of most solids in liquids.

ANSWER 13

Solubility is the amount of solute that dissolves in a specified amount of solvent at a stated temperature (and pressure for gases) to form a saturated solution. Most solids become more soluble as temperature rises, but exceptions occur.

QUESTION 14 [M] State Henry's law and write its mathematical expression.

ANSWER 14

Henry's law: at fixed temperature, dissolved-gas mole fraction is proportional to its partial pressure: $p = k_H x$.

QUESTION 15 [H] Explain why the solubility of a gas in a liquid generally decreases when temperature increases.

ANSWER 15

Gas dissolution is usually exothermic. Heating favours gas escape (Le Châtelier), so gas solubility generally falls as temperature rises.

QUESTION 16 [M] Why are soft drinks bottled under high pressure?

ANSWER 16

High CO_2 pressure increases its dissolved mole fraction by Henry's law; opening the bottle lowers pressure and the gas escapes as bubbles.

QUESTION 17 [H] At a fixed temperature, explain how the mole fraction of a dissolved gas depends on its partial pressure.

ANSWER 17

At fixed temperature, Henry's law gives $x = p/k_H$; dissolved-gas mole fraction rises linearly with partial pressure in the dilute range.

QUESTION 18 [M] What are solid solutions? Give one example of a substitutional solid solution.

ANSWER 18

A solid solution is a homogeneous solid mixture. Example: substitutional Cu–Ni alloy, where atoms of one metal replace atoms of the other in the lattice.

QUESTION 19 [M] Define colligative properties and list the four principal colligative properties.

ANSWER 19

Colligative properties depend on the number of dissolved particles, not their identity. The four are relative lowering of vapour pressure, boiling-point elevation, freezing-point depression and osmotic pressure.

QUESTION 20 [H] Why do colligative properties depend on the number of solute particles rather than their chemical identity?

ANSWER 20

In dilute ideal solutions, solvent chemical potential changes according to solute-particle mole fraction. Therefore effects scale with particle count; dissociation/association changes the count and is captured by i.

QUESTION 21 [M] Define relative lowering of vapour pressure.

ANSWER 21

Relative lowering is $(p^\circ - p)/p^\circ$, where p° is pure-solvent vapour pressure and p is solution vapour pressure. For an ideal solution with non-volatile solute it equals x_{solute}.

QUESTION 22 [M] Define elevation in boiling point and write its relation to molality.

ANSWER 22

Boiling-point elevation is $\Delta T_b = T_b(\text{solution}) - T_b^\circ$; for a dilute solution, $\Delta T_b = K_b\, m$ (or $iK_b m$ for an electrolyte).

QUESTION 23 [M] Define depression in freezing point and write its relation to molality.

ANSWER 23

Freezing-point depression is $\Delta T_f = T_f^\circ - T_f(\text{solution})$; for a dilute solution, $\Delta T_f = K_f\, m$ (or $iK_f m$ for an electrolyte).

QUESTION 24 [M] What is osmotic pressure?

ANSWER 24

Osmotic pressure π is the minimum external pressure applied to a solution to stop net solvent flow through a semipermeable membrane.

QUESTION 25 [H] Derive the equation $\pi = CRT$ for a dilute solution and state the assumptions involved.

ANSWER 25

At equilibrium, $\mu_A^*(P) = \mu_A(\text{solution}, P+\pi) \approx \mu_A^*(P) + V_A\pi + RT \ln x_A$, so $\pi = -(RT/V_A)\ln x_A$. For a dilute solution, $-\ln x_A \approx x_B \approx n_B/n_A$ and $V \approx n_A V_A$; hence $\pi V \approx n_B RT$, or $\pi = CRT$ with $C = n_B/V$. Assumptions: ideal dilute solution, small pressure correction, constant solvent partial molar volume, solvent-only membrane, uniform T, and negligible solute volatility.

QUESTION 26 [M] Why is osmotic pressure useful for determining the molar mass of macromolecules?

ANSWER 26

For dilute polymer solutions, π is measurable at low concentration without heating or freezing the macromolecule. From $\pi V = nRT$, its molar mass can be obtained even when other colligative changes are very small.

QUESTION 27 [H] Compare reverse osmosis and osmosis. Mention one practical application of reverse osmosis.

ANSWER 27

Osmosis is net solvent flow from lower solute concentration (lower π) to higher π through a semipermeable membrane. Reverse osmosis applies pressure greater than π to reverse that flow; it is used for desalination.

QUESTION 28 [M] What are isotonic, hypertonic and hypotonic solutions?

ANSWER 28

Isotonic solutions have equal osmotic pressure at the same T. Hypertonic has higher π than the reference; hypotonic has lower π. Solvent tends to flow toward the hypertonic side.

QUESTION 29 [H] Predict the direction of solvent flow when two solutions of different osmotic pressures are separated by a semipermeable membrane.

ANSWER 29

Solvent flows from the solution of lower osmotic pressure to that of higher osmotic pressure until equilibrium, provided the membrane is selectively permeable to solvent.

QUESTION 30 [M] What is an abnormal molar mass?

ANSWER 30

An abnormal molar mass is an experimentally inferred value that differs from the formula molar mass because solute particles associate or dissociate in solution.

QUESTION 31 [M] Define the van’t Hoff factor and write its relation with observed and calculated colligative properties.

ANSWER 31

*The van’t Hoff factor i is observed colligative effect divided by the value calculated for a non-associating, non-dissociating solute at the same concentration. Equivalently, i=M*_formula/*M*_observed.

QUESTION 32 [H] Explain how association of solute particles affects the van’t Hoff factor and observed molar mass.

ANSWER 32

Association reduces the number of solute particles, so $i<1$. Since $M_{observed}=M_{formula}/i$, the apparent molar mass is greater than the formula molar mass.

QUESTION 33 [H] Explain how dissociation of a solute affects its van’t Hoff factor and observed molar mass.

ANSWER 33

Dissociation increases particle count, so $i>1$. Consequently $M_{observed}=M_{formula}/i$ is smaller than the formula molar mass.

QUESTION 34 [M] Why does NaCl show a van’t Hoff factor greater than one in a dilute aqueous solution?

ANSWER 34

In dilute water NaCl dissociates mainly into Na^+ and Cl^-, increasing the number of particles; hence i is greater than one (approaching 2 at ideal infinite dilution).

QUESTION 35 [H] For a solute that dimerises, derive the relation between degree of association and van’t Hoff factor.

ANSWER 35

For initial N molecules, dimerisation of fraction α leaves $N(1-\alpha)+N\alpha/2$ particles. Thus $i=1-\alpha/2$.

QUESTION 36 [H] For a solute that dissociates into n particles, derive the relation between degree of dissociation and van’t Hoff factor.

ANSWER 36

If one solute unit produces n particles upon dissociation, initial N units yield $N(1-\alpha)+n N\alpha$ particles. Therefore $i=1+\alpha(n-1)$.

QUESTION 37 [M] A solution contains 2 g of a non-volatile solute in 100 g water. What data are needed to calculate its molar mass using freezing-point depression?

ANSWER 37

Required data: solute mass, solvent mass, K_f, measured ΔT_f, and confirmation that $i=1$ (or an independently known i). Then $m=\Delta T_f/(iK_f)$ and molar mass=solute mass/(molality×kg solvent).

QUESTION 38 [H] Calculate the freezing-point depression for a 0.20 mol kg^{-1} aqueous solution of a non-electrolyte. Kf for water is 1.86 K kg mol^{-1}.

ANSWER 38 · CORRECTED

For a non-electrolyte, $\Delta T_f=K_f m=(1.86\ K\ kg\ mol^{-1})(0.20\ mol\ kg^{-1})=0.372\ K$ unrounded; to 2 significant figures (limited by 0.20 m), $\Delta T_f=0.37\ K$.

QUESTION 39 [H] A 0.1 mol kg^{-1} solution of a solute has ΔTf = 0.279 K in water. Calculate its van't Hoff factor. Kf = 1.86 K kg mol^{-1}.

ANSWER 39 · NEEDS QUALIFICATION

$i=\Delta T_f/(K_f m)=0.279/(1.86\times0.10)=1.50$ unrounded. If 0.10 m is intended (2 significant figures), report $i\approx1.5$; if the printed 0.1 m is strictly 1 significant figure, i rounds coarsely to 2.

QUESTION 40 [H] A solution contains 1.0 g of a polymer in 100 mL solution and has an osmotic pressure of 0.0246 atm at 300 K. Estimate its molar mass using R = 0.0821 L atm mol^{-1} K^{-1}.

ANSWER 40 · CORRECTED

$n=\pi V/RT=(0.0246\ atm\times0.100\ L)/(0.0821\ L\ atm\ mol^{-1}\ K^{-1}\times300\ K)=9.99\times10^{-5}$ mol. $M=1.0\ g/n\approx1.00\times10^{4}\ g\ mol^{-1}$ unrounded; to 2 significant figures (limited by 1.0 g), $M\approx1.0\times10^{4}\ g\ mol^{-1}$. The trailing-zero precision of 100 mL and 300 K is also unspecified.

QUESTION 41 [M] Why is the boiling point of a solution containing a non-volatile solute higher than that of the pure solvent?

ANSWER 41

A non-volatile solute lowers solvent vapour pressure. The solution must be heated to a higher temperature to reach the external pressure at boiling.

QUESTION 42 [H] Explain why adding ethylene glycol to water lowers the freezing point of the mixture.

ANSWER 42

Ethylene glycol lowers the mole fraction and chemical potential of water; the liquid must be cooled further to reach equilibrium with ice, so its freezing point falls.

QUESTION 43 [M] State the relationship between elevation in boiling point and depression in freezing point for the same molality.

ANSWER 43

For equal molality in the same solvent, $\Delta T_b/\Delta T_f = K_b/K_f$. The two temperature changes are not generally equal because the solvent constants differ.

QUESTION 44 [H] A solution has a higher vapour pressure than predicted by Raoult's law. What can you infer about solute–solvent interactions?

ANSWER 44

Positive deviation implies weaker solute–solvent attractions than the average like-component attractions; molecules escape more readily and vapour pressure is higher.

QUESTION 45 [M] Distinguish between ideal and non-ideal solutions.

ANSWER 45

Ideal solutions obey Raoult's law throughout and have $\Delta H_{mix} = \Delta V_{mix} = 0$. Non-ideal solutions deviate because unlike interactions differ; ΔH_{mix} and/or ΔV_{mix} are non-zero.

QUESTION 46 [H] Explain why a mixture of ethanol and acetone may show positive deviation from Raoult's law.

ANSWER 46

Mixing acetone disrupts some ethanol–ethanol hydrogen bonding, and acetone–ethanol attractions are comparatively weaker than the original interactions; this can produce positive deviation.

QUESTION 47 [M] What is an azeotrope? Distinguish between minimum-boiling and maximum-boiling azeotropes.

ANSWER 47

An azeotrope boils at fixed composition because liquid and vapour compositions match, so it distils without composition change. Positive deviation gives a minimum-boiling azeotrope; negative deviation gives a maximum-boiling one.

QUESTION 48 [H] Explain why the composition of an azeotrope cannot be separated completely by simple fractional distillation.

ANSWER 48

At the azeotropic composition, vapour has the same composition as liquid. Fractional distillation cannot enrich one component past that composition at the azeotropic pressure.

QUESTION 49 [M] Why is the vapour pressure of a pure liquid affected by temperature?

ANSWER 49

Increasing temperature raises the fraction of molecules with enough energy to escape the liquid, so equilibrium vapour pressure increases.

QUESTION 50 [H] A solution is prepared by mixing equal masses of two liquids with different molar masses. Explain why their mole fractions will not generally be equal.

ANSWER 50

Mole fraction is based on moles, not mass. For equal masses, $n=m/M$; the liquid with lower molar mass contributes more moles, so the fractions generally differ.

QUESTION 51 [M] Calculate the mass percentage of solute in a solution made by dissolving 12 g solute in 108 g solvent.

ANSWER 51 · CORRECTED

Mass percent $=100\times12/(12+108)=10.0\%$ (w/w) by calculation; to the precision implied by the stated masses, report approximately 10%.

QUESTION 52 [H] A 2 molal aqueous solution is prepared. Calculate the moles of solute present per 500 g of water.

ANSWER 52 · NEEDS QUALIFICATION

$n_{solute} = (2\ mol\ kg^{-1})(0.500\ kg\ water) = 1.0\ mol$ by calculation; to 1 significant figure from the stated "2 molal," report 1 mol. If "2.0 molal" is intended, report 1.00 mol.

QUESTION 53 [M] Convert a solution containing 10% (w/w) solute into a description of what the percentage means.

ANSWER 53

10% (w/w) means 10 g solute in every 100 g of total solution (90 g solvent for a binary solution).

QUESTION 54 [H] A solution contains 5 g urea in 95 g water. Calculate its molality.

ANSWER 54 · CORRECTED

$n(urea) = 5/60.06 = 0.08325\ mol$; solvent = 0.095 kg.
Molality $= 0.08325/0.095 = 0.8763\ mol\ kg^{-1}$ ($\approx 0.88\ mol\ kg^{-1}$ if 5 g is intended to 2 significant figures; strictly, the unqualified "5 g" indicates 1 significant figure and would round to 0.9 mol kg^{-1}).

QUESTION 55 [M] Explain why the vapour pressure of a solution changes when the mole fraction of a volatile component changes.

ANSWER 55

For each volatile component, $p_i = x_i\ p_i^\circ$ in an ideal solution. Changing its mole fraction changes its partial pressure and therefore the total vapour pressure.

QUESTION 56 [H] For an ideal binary solution, write expressions for total vapour pressure and partial vapour pressures.

ANSWER 56

For ideal binary liquids A and B: $p_A = x_A p_A^\circ$, $p_B = x_B p_B^\circ$; $P_{total} = x_A p_A^\circ + x_B p_B^\circ$.

QUESTION 57 [M] What is the significance of the slope of a vapour-pressure versus mole- fraction plot for an ideal solution?

ANSWER 57

For P_{total} versus x_A, slope is $p_A^\circ - p_B^\circ$. Partial-pressure lines have slopes $+p_A^\circ$ for A and $-p_B^\circ$ for B when plotted against x_A.

QUESTION 58 [H] Explain how intermolecular interactions determine whether mixing produces positive or negative deviation.

ANSWER 58

If unlike A–B attractions are weaker, molecules escape more readily (positive deviation); if stronger, escape is suppressed (negative deviation). Heat/volume changes often have matching signs but are not the defining criterion.

QUESTION 59 [M] Why does a gas dissolve more readily in a liquid at higher pressure?

ANSWER 59

Higher pressure increases gas chemical potential above the liquid and drives more gas into solution; at fixed T the proportionality is Henry's law.

QUESTION 60 [H] A gas has Henry's law constant $k_H = 1.0 \times 10^5$ kPa and partial pressure 50 kPa. Calculate its mole fraction in solution using $p = k_H x$.

ANSWER 60 · NEEDS QUALIFICATION

$x = p/k_H = 50\ \text{kPa}/(1.0\times10^5\ \text{kPa}) = 5.0\times10^{-4}$ (or 5×10^{-4} if 50 kPa is intended to 1 significant figure; the trailing zero is ambiguous).

QUESTION 61 [M] Distinguish between molality and mole fraction.

ANSWER 61

Molality is moles solute per kilogram solvent and depends on solvent mass. Mole fraction is moles of one component divided by total moles and is dimensionless.

QUESTION 62 [H] Calculate the boiling-point elevation of a 0.50 m glucose solution in water. $K_b = 0.512$ K kg mol^{-1}.

ANSWER 62 · CORRECTED

$\Delta T_b = K_b m = (0.512\ \text{K kg mol}^{-1})(0.50\ \text{mol kg}^{-1}) = 0.256$ K unrounded; to 2 significant figures (limited by 0.50 m), $\Delta T_b = 0.26$ K.

QUESTION 63 [M] Why is a semipermeable membrane essential for osmotic pressure measurements?

ANSWER 63

A semipermeable membrane must pass solvent but retain solute; otherwise no selective solvent flow or measurable osmotic pressure develops.

QUESTION 64 [H] A 0.05 M solution has osmotic pressure 1.23 atm at 300 K. Calculate its van’t Hoff factor using R = 0.0821 L atm mol^{-1} K^{-1}.

ANSWER 64 · NEEDS QUALIFICATION

$i=\pi/(CRT)=1.23/[0.050\times0.0821\times300]=0.9988\approx1.0$; the stated concentration is only 0.05 M, so the result is appropriately interpreted as about unity.

QUESTION 65 [M] State one limitation of using boiling-point elevation to determine molar mass.

ANSWER 65

It requires precise temperature measurement and enough sample; small ΔT_b may be difficult to resolve. Volatile or thermally unstable solutes/solvents can also invalidate the method.

QUESTION 66 [H] A solute dissociates into three ions and has degree of dissociation 0.60. Calculate its van’t Hoff factor.

ANSWER 66 · CORRECTED

For dissociation into 3 particles, $i=1+\alpha(n-1)=1+0.60(2)=2.2$ (2 significant figures).

QUESTION 67 [M] What does a van’t Hoff factor equal to one indicate in an ideal dilute solution?

ANSWER 67

$i=1$ indicates the effective particle count equals the formula-unit count: negligible association/dissociation in the ideal dilute limit.

QUESTION 68 [H] A compound associates to form trimers. If the degree of association is α, derive the expression for i.

ANSWER 68

For trimerisation, $i=1-2\alpha/3$ because each associated group of three replaces three particles by one.

QUESTION 69 [M] Why is the freezing-point depression method unsuitable for some thermally unstable solutes?

ANSWER 69

The method requires heating to boiling; a thermally unstable solute may decompose or react, so the measured boiling point would not represent the original solution.

QUESTION 70 [H] Explain how experimental molar mass differing from formula molar mass can reveal association or dissociation.

ANSWER 70

Association gives $i<1$ and $M_{observed}>M_{formula}$; dissociation gives $i>1$ and $M_{observed}<M_{formula}$. The direction and size of the discrepancy diagnose particle-number change.

QUESTION 71 [M] A solution contains a volatile solute and a volatile solvent. How does each component contribute to total vapour pressure?

ANSWER 71

Each volatile component contributes a partial pressure; for an ideal mixture $p_i = x_i p_i^\circ$, and Dalton's law gives $P_{total} = \sum_i p_i$.

QUESTION 72 [H] A 0.20 m solution of $CaCl_2$ is assumed to dissociate completely. Calculate its boiling-point elevation in water using Kb = 0.512 K kg mol^{-1}.

ANSWER 72 · CORRECTED

$CaCl_2 \rightarrow Ca^{2+} + 2Cl^-$, so $i \approx 3$. $\Delta T_b = iK_b m = 3 \times 0.512 \times 0.20 = 0.3072$ K unrounded; to 2 significant figures (limited by 0.20 m), $\Delta T_b \approx 0.31$ K (complete dissociation, dilute-solution assumption).

UNIT 2: ELECTROCHEMISTRY

Questions 73–144 · 72 complete pairs

QUESTION 73 [M] Define oxidation and reduction in terms of electron transfer.

ANSWER 73

Oxidation is loss of electrons; reduction is gain of electrons. They occur together in a redox reaction.

QUESTION 74 [M] Distinguish between a galvanic cell and an electrolytic cell.

ANSWER 74

A galvanic cell uses a spontaneous redox reaction to produce electrical energy (anode negative); an electrolytic cell uses an external supply to drive a non-spontaneous reaction (anode positive). Oxidation is at the anode and reduction at the cathode in both.

QUESTION 75 [M] What is the EMF of an electrochemical cell?

ANSWER 75

EMF is the open-circuit potential difference between the cell terminals, equal to the maximum electrical work per unit charge under specified conditions.

QUESTION 76 [M] Define standard electrode potential and state the standard conditions.

ANSWER 76

Standard electrode potential is a reduction potential measured relative to the standard hydrogen electrode under standard conditions: solute activities 1 (approximately 1 M), gases 1 bar, pure solids/liquids, specified temperature (usually 298 K).

QUESTION 77 [H] Explain why the standard hydrogen electrode is assigned a potential of zero volts.

ANSWER 77

An absolute single-electrode potential cannot be measured independently; only cell differences are measurable. Assigning the SHE zero is a reference convention that allows all other electrode potentials to be tabulated consistently.

QUESTION 78 [M] Write the cell notation for a Daniell cell.

ANSWER 78

Daniell cell: $Zn(s)|Zn^{2+}(aq, 1\ M)||Cu^{2+}(aq, 1\ M)|Cu(s)$, with the anode on the left.

QUESTION 79 [H] Derive the relation between standard cell potential and standard Gibbs energy change.

ANSWER 79

Maximum non-expansion electrical work is $-nFE_{cell}$. At reversible standard conditions $\Delta G^\circ = w_{max}$, so $\Delta G^\circ = -nFE^\circ_{cell}$.

QUESTION 80 [M] State the Nernst equation for a general cell reaction.

ANSWER 80

For a balanced cell reaction, $E = E^\circ - (RT/nF)\ln Q$. At 298 K, $E = E^\circ - (0.05916\ V/n)\log_{10} Q$.

QUESTION 81 [H] Write the Nernst equation for the cell $Zn(s)|Zn^{2+}(aq)||Cu^{2+}(aq)|Cu(s)$ at 298 K.

ANSWER 81

$Zn(s)+Cu^{2+}\rightarrow Zn^{2+}+Cu(s)$, $Q=a(Zn^{2+})/a(Cu^{2+})$. $E=E°-(RT/2F)\ln([Zn^{2+}]/[Cu^{2+}])$; at 298 K use 0.05916/2 times $\log_{10}$ of that ratio.

QUESTION 82 [M] What is the relationship between cell potential and spontaneity of a cell reaction?

ANSWER 82

$\Delta G=-nFE_{cell}$: $E>0$ means $\Delta G<0$ and forward reaction spontaneous; $E=0$ is equilibrium; $E<0$ means the forward reaction is non-spontaneous as written.

QUESTION 83 [H] A cell reaction has n = 2 and E°cell = 1.10 V. Calculate ΔG° using F = 96500 C mol^{-1}.

ANSWER 83 · CORRECTED

$\Delta G°=-nFE°=-(2)(96500)(1.10)=-2.123\times10^{5}$ J $mol^{-1}=-212.3$ kJ mol^{-1} unrounded; to 3 significant figures (from $E°=1.10$ V), $\Delta G°=-212$ kJ mol^{-1}.

QUESTION 84 [M] Define the equilibrium constant of a cell reaction.

ANSWER 84

For the balanced cell reaction, K is the equilibrium ratio of product activities to reactant activities, each raised to its stoichiometric coefficient; pure solids and liquids are omitted.

QUESTION 85 [H] Derive the relation between E°cell and the equilibrium constant at 298 K.

ANSWER 85

At equilibrium $E=0$ and $Q=K$. The Nernst equation gives $E°=(RT/nF)\ln K$; at 298 K, $E°=(0.05916\ V/n)\log_{10}K$.

QUESTION 86 [M] What is the function of a salt bridge in a galvanic cell?

ANSWER 86

The salt bridge completes the ionic circuit and maintains electroneutrality by ion migration, while limiting direct mixing of the half-cell solutions.

QUESTION 87 [H] Explain why a galvanic cell stops producing current when its cell reaction reaches equilibrium.

ANSWER 87

At equilibrium $Q=K$, so $E=0$ and $\Delta G=0$. There is no net thermodynamic driving force for electron flow or net cell reaction.

QUESTION 88 [M] Define specific conductivity and state its SI unit.

ANSWER 88

Specific conductivity κ is conductance of a solution column of unit length and unit cross-sectional area ($\kappa = G\,l/A$). SI unit: $S\ m^{-1}$.

QUESTION 89 [M] Define molar conductivity and state its usual unit.

ANSWER 89

Molar conductivity Λ_m is conductivity multiplied by volume containing one mole of electrolyte. Usual unit: $S\ cm^2\ mol^{-1}$ (SI: $S\ m^2\ mol^{-1}$).

QUESTION 90 [H] Explain why conductivity decreases on dilution while molar conductivity generally increases.

ANSWER 90

Dilution reduces the number of ions per unit volume, so κ falls. Per mole, ions experience less interionic restraint and weak electrolytes ionise more, so Λ_m rises.

QUESTION 91 [M] How does molar conductivity vary with concentration for strong and weak electrolytes?

ANSWER 91

For strong electrolytes Λ_m rises modestly on dilution as interionic effects diminish; for weak electrolytes it rises sharply because dilution also increases ionisation.

QUESTION 92 [H] Explain why the increase in molar conductivity on dilution is much greater for weak electrolytes.

ANSWER 92

A weak electrolyte is only partly ionised when concentrated; dilution shifts its ionisation equilibrium toward ions, increasing the number of charge carriers and Λ_m substantially.

QUESTION 93 [M] State Kohlrausch’s law of independent migration of ions.

ANSWER 93

At infinite dilution each ion contributes independently to total molar conductivity: $\Lambda_m^\circ = \nu_+\lambda_+^\circ + \nu_-\lambda_-^\circ$, with stoichiometric coefficients ν_i.

QUESTION 94 [H] Show how Kohlrausch’s law can be used to determine the limiting molar conductivity of a weak electrolyte.

ANSWER 94

By ionic additivity, $\Lambda_m^\circ(CH_3COOH) = \Lambda_m^\circ(HCl) + \Lambda_m^\circ(CH_3COONa) - \Lambda_m^\circ(NaCl)$; the strong-electrolyte terms cancel Na^+ and Cl^-, leaving H^+ and acetate contributions.

QUESTION 95 [M] How can the degree of dissociation of a weak electrolyte be obtained from conductivity data?

ANSWER 95

For a weak electrolyte, degree of dissociation $\alpha = \Lambda_m/\Lambda_m^\circ$, where Λ_m is measured at the concentration and Λ_m° is the limiting value.

QUESTION 96 [H] Explain how the dissociation constant of a weak electrolyte can be calculated using conductivity measurements.

ANSWER 96

Measure Λ_m and obtain Λ_m° (e.g. by Kohlrausch’s law); set $\alpha = \Lambda_m/\Lambda_m^\circ$. For a weak 1:1 electrolyte of concentration c, $K_a = c\alpha^2/(1-\alpha)$.

QUESTION 97 [M] State Faraday’s first law of electrolysis.

ANSWER 97

Faraday’s first law: the mass deposited or liberated at an electrode is directly proportional to the charge passed, $m \propto Q$.

QUESTION 98 [M] State Faraday’s second law of electrolysis.

ANSWER 98

Faraday’s second law: for the same charge through different electrolytes, deposited masses are proportional to their chemical equivalent masses (molar mass divided by electron number).

QUESTION 99 [H] Calculate the charge required to deposit one mole of silver from Ag^+ ions.

ANSWER 99

$Ag^+ + e^- \rightarrow Ag$ requires one mole of electrons per mole Ag. $Q = F = 9.65 \times 10^4$ C to deposit one mole.

QUESTION 100 [H] How much copper is deposited by passing 2.0 A current for 30 minutes through $CuSO_4$ solution? Use F = 96500 C mol^{-1} and Cu = 63.5 g mol^{-1}.

ANSWER 100 · CORRECTED

$Q = It = 2.0 \times (30 \times 60) = 3600$ C. $n(Cu) = Q/(2F) = 0.01865$ mol; $m = 0.01865 \times 63.5 = 1.18$ g unrounded; to 2 significant figures (from 2.0 A), $m \approx 1.2$ g. The precision of the stated 30 min is ambiguous; if it is 1 significant figure, strict final reporting is about 1 g.

QUESTION 101 [M] What is the role of the electrolyte in an electrolytic cell?

ANSWER 101

The electrolyte supplies mobile ions to carry current, completes ionic conduction, and undergoes the electrode reactions; it must be sufficiently ionically conducting.

QUESTION 102 [H] Compare the products of electrolysis of aqueous NaCl and molten NaCl.

ANSWER 102 · NEEDS QUALIFICATION

Molten NaCl: Na^+ is reduced to Na(l) at the cathode; Cl^- is oxidised to $Cl_2(g)$ at the anode. Aqueous NaCl: water is reduced to H_2 at the cathode and NaOH remains; concentrated brine gives Cl_2 at the anode, while very dilute chloride can give O_2 instead.

QUESTION 103 [M] Describe the construction and working principle of a dry cell.

ANSWER 103 · CORRECTED

A Leclanché dry cell has a zinc container (anode), a central carbon rod surrounded by MnO_2/carbon (cathode mixture), and moist $NH_4Cl/ZnCl_2$ paste. Approximate discharge reactions are $Zn \rightarrow Zn^{2+} + 2e^-$ and $MnO_2 + NH_4^+ + e^- \rightarrow MnO(OH) + NH_3$; the cathodic depolariser consumes electrons. It is a primary, non-rechargeable cell.

QUESTION 104 [M] Describe the main components of a lead accumulator.

ANSWER 104

A lead accumulator has a Pb negative plate, PbO_2 positive plate, porous separators and aqueous H_2SO_4 electrolyte in a container. Both plates become $PbSO_4$ during discharge; charging reverses the reaction.

QUESTION 105 [H] Write the electrode reactions occurring during discharge of a lead accumulator.

ANSWER 105

Negative plate: $Pb + SO_4^{2-} \rightarrow PbSO_4 + 2e^-$. Positive plate: $PbO_2 + 4H^+ + SO_4^{2-} + 2e^- \rightarrow PbSO_4 + 2H_2O$. Overall: $Pb + PbO_2 + 2H_2SO_4 \rightarrow 2PbSO_4 + 2H_2O$.

QUESTION 106 [M] What is a fuel cell? Give one advantage over a conventional cell.

ANSWER 106

A fuel cell continuously converts supplied fuel and oxidant directly to electrical energy through redox reactions. An advantage is high efficiency with low local pollutant emission (for H_2/O_2, water is the product).

QUESTION 107 [H] Write the anode, cathode and overall reactions for a hydrogen–oxygen fuel cell in alkaline medium.

ANSWER 107

In alkaline electrolyte: anode $H_2 + 2OH^- \rightarrow 2H_2O + 2e^-$; cathode $\frac{1}{2}O_2 + H_2O + 2e^- \rightarrow 2OH^-$. Overall $H_2 + \frac{1}{2}O_2 \rightarrow H_2O$.

QUESTION 108 [M] Why is corrosion considered an electrochemical process?

ANSWER 108

Corrosion involves spatially separated anodic metal oxidation and cathodic reduction reactions on a conducting wet surface, coupled by electron and ion transport—an electrochemical cell.

QUESTION 109 [H] Explain the electrochemical mechanism of rusting of iron.

ANSWER 109

At anodic sites $Fe \rightarrow Fe^{2+} + 2e^-$. At aerated cathodic sites $O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$. Fe^{2+} forms $Fe(OH)_2$, then oxidises/hydrates to Fe(III) oxyhydroxides and hydrated rust, often represented as $Fe_2O_3 \cdot xH_2O$.

QUESTION 110 [M] What is the difference between primary and secondary batteries?

ANSWER 110

Primary batteries are normally non-rechargeable because their cell reaction is not practically reversible. Secondary batteries can be recharged by driving the reverse reaction with an external current.

QUESTION 111 [H] A cell has E°cell = 0.46 V. Explain how the sign of ΔG° is determined and what it implies.

ANSWER 111

$\Delta G^\circ = -nFE^\circ$. Since $E^\circ = +0.46$ V and $n > 0$, $\Delta G^\circ < 0$; the reaction as written is standard-state spontaneous.

QUESTION 112 [M] Define the electrochemical series and mention two applications.

ANSWER 112

The electrochemical series ranks redox couples by standard reduction potential. It helps predict cell EMFs/spontaneity and compare oxidising or reducing strengths, including likely displacement reactions.

QUESTION 113 [H] Use standard electrode potentials to explain how to predict whether one metal can displace another from its salt solution.

ANSWER 113

Compare standard reduction potentials: the species with the more positive E° is reduced at the cathode. A metal M can displace N^{2+} if $E^\circ_{cell} = E^\circ(N^{2+}/N) - E^\circ(M^{2+}/M) > 0$ for $M + N^{2+} \rightarrow M^{2+} + N$.

QUESTION 114 [M] What is the standard electrode potential of an electrode measured against?

ANSWER 114

It is measured relative to the standard hydrogen electrode (SHE), whose potential is defined as zero under standard conditions.

QUESTION 115 [H] Explain why Ecell changes as the concentrations of ions change during cell operation.

ANSWER 115

As reaction proceeds, ion activities change and hence Q changes. The Nernst equation $E = E^\circ - (RT/nF)\ln Q$ therefore makes the cell potential vary until it reaches zero at equilibrium.

QUESTION 116 [M] Define limiting molar conductivity.

ANSWER 116

Limiting molar conductivity Λ_m° is the molar conductivity extrapolated to infinite dilution, where interionic interactions vanish and ions migrate independently.

QUESTION 117 [H] Explain the difference between ionic mobility and ionic conductivity.

ANSWER 117

Ionic mobility is drift velocity per unit electric field for a particular ion. Conductivity is the bulk ability of all ions in solution to carry current and depends on concentration, charge and mobility.

QUESTION 118 [M] Why does temperature generally increase the conductivity of an electrolytic solution?

ANSWER 118

Higher temperature generally lowers solvent viscosity and increases ionic mobility; it may also increase weak-electrolyte ionisation, raising conductivity.

QUESTION 119 [H] A conductivity cell has resistance 200 Ω and cell constant 1.25 cm^{-1}. Calculate conductivity in S cm^{-1}.

ANSWER 119 · NEEDS QUALIFICATION

κ=cell constant/R=1.25 cm^{-1}/200 Ω=0.00625 S cm^{-1} unrounded. The "200 Ω" precision is ambiguous: report 0.006 S cm^{-1}, 0.0063 S cm^{-1}, or 0.00625 S cm^{-1} if 200 has 1, 2, or 3 significant figures, respectively.

QUESTION 120 [H] A 0.01 M electrolyte solution has conductivity 1.5×10^{-3} S cm^{-1}. Calculate its molar conductivity in S cm^2 mol^{-1}.

ANSWER 120 · NEEDS QUALIFICATION

$\Lambda_m=\kappa\times1000/c=(1.5\times10^{-3}$ S $cm^{-1}\times1000$ cm^3 $L^{-1})/(0.01$ mol $L^{-1})=1.5\times10^2$ S cm^2 mol^{-1} unrounded. Because 0.01 M has 1 significant figure as written, strict rounding gives 2×10^2 S cm^2 mol^{-1}; 1.5×10^2 applies if 0.01 is intended to 2 significant figures.

QUESTION 121 [M] Explain why electrolytic conduction differs from metallic conduction.

ANSWER 121

Metallic conduction is electron motion without bulk chemical change. Electrolytic conduction is ion motion in solution or melt and is accompanied by electrode redox reactions.

QUESTION 122 [H] A reaction has E°cell = 0.00 V. What can be inferred about ΔG° and the equilibrium constant at standard conditions?

ANSWER 122

For $n>0$, $\Delta G°=-nFE°=0$. Also $\Delta G°=-RT\ \ln K$, hence $K=1$ at that temperature under standard-state conventions.

QUESTION 123 [M] What is the direction of electron flow in an operating Daniell cell?

ANSWER 123

Electrons flow through the external circuit from the zinc anode to the copper cathode.

QUESTION 124 [H] Given $E°(Cu^{2+}/Cu) = +0.34$ V and $E°(Zn^{2+}/Zn) = -0.76$ V, calculate $E°_{cell}$ for a Daniell cell.

ANSWER 124

$E°_{cell} = E°_{cathode} - E°_{anode} = 0.34 - (-0.76) = 1.10$ V.

QUESTION 125 [M] Why is the anode negative in a galvanic cell but positive in an electrolytic cell?

ANSWER 125

Anode is where oxidation supplies electrons in a galvanic cell, making it negative. In an electrolytic cell the power supply removes electrons from the anode, making it positive; oxidation still occurs there.

QUESTION 126 [H] For a cell reaction involving two electrons, write $\Delta G°$ in terms of $E°_{cell}$ and Faraday constant.

ANSWER 126

For a two-electron cell reaction, $\Delta G° = -2FE°_{cell}$ (per mole of reaction as written).

QUESTION 127 [M] How does a salt bridge maintain electrical neutrality in the two half- cells?

ANSWER 127

Anions from the bridge migrate toward the anode compartment, where positive charge is produced; cations migrate toward the cathode compartment, where cations are consumed, maintaining bulk neutrality.

QUESTION 128 [H] Use the Nernst equation to predict how Ecell changes when the concentration of the oxidised species increases.

ANSWER 128

For $Ox + ne^- \rightarrow Red$, $Q = a(Red)/a(Ox)$; increasing Ox lowers Q and raises E by Nernst. If the oxidised species occurs on the product side of the chosen cell reaction, the direction reverses; the balanced reaction must be specified.

QUESTION 129 [M] What is the reaction quotient Q in the Nernst equation?

ANSWER 129

Q is the ratio of instantaneous product activities to reactant activities, each raised to its stoichiometric coefficient; pure solids and liquids are omitted. Unlike K, it need not describe equilibrium.

QUESTION 130 [H] For a cell reaction with Q = 100, n = 2 and E°cell = 0.50 V at 298 K, calculate Ecell.

ANSWER 130 · CORRECTED

$E = E° - (0.05916/2)\log_{10}(100) = 0.50 - 0.05916 = 0.44084$ V unrounded; to 2 significant figures (from E°=0.50 V), E≈0.44 V.

QUESTION 131 [M] Why is molar conductivity reported per mole of electrolyte rather than per unit volume?

ANSWER 131

Molar conductivity normalises conductivity to the amount of electrolyte, allowing comparison of ion-transport contribution per mole at different concentrations.

QUESTION 132 [H] Explain how ionic inter-attraction affects molar conductivity at finite concentration.

ANSWER 132

At finite concentration, oppositely charged ions attract and like-charged ions repel, hindering independent motion and lowering ionic mobility and Λ_m; dilution reduces these effects.

QUESTION 133 [M] What is cell constant and how is it determined?

ANSWER 133

Cell constant is electrode separation divided by electrode area, l/A (cm^{-1}). Determine it by measuring resistance/conductance of a standard solution with known κ.

QUESTION 134 [H] A conductivity cell has conductance 0.005 S and cell constant 0.8 cm^{-1}. Calculate conductivity.

ANSWER 134 · CORRECTED

$\kappa = G \times \text{cell constant} = (0.005\ S)(0.8\ cm^{-1}) = 4 \times 10^{-3}\ S\ cm^{-1}$ (1 significant figure).

QUESTION 135 [M] Why does a weak electrolyte ionise more on dilution?

ANSWER 135

For a weak electrolyte, ionisation is an equilibrium; dilution favours the side with more dissolved particles, increasing the fraction ionised (Ostwald dilution law).

QUESTION 136 [H] Use Kohlrausch's law to express $\Lambda°_m(CH_3COOH)$ using limiting ionic conductivities of suitable strong electrolytes.

ANSWER 136

$\Lambda°_m(CH_3COOH) = \Lambda°_m(HCl) + \Lambda°_m(CH_3COONa) - \Lambda°_m(NaCl)$.

QUESTION 137 [M] What is electrochemical equivalent?

ANSWER 137

Electrochemical equivalent Z is mass deposited/liberated per unit charge passed: $Z = m/Q$; SI unit kg C^{-1}.

QUESTION 138 [H] Calculate the time needed to deposit 1.08 g Ag using a 1.0 A current. Ag = 108 g mol^{-1}; F = 96500 C mol^{-1}.

ANSWER 138 · CORRECTED

$n(Ag) = 1.08/108 = 0.0100$ mol, so $Q = 0.0100F = 965$ C. At 1.0 A, $t = Q/I = 965$ s unrounded; to 2 significant figures (from 1.0 A), $t \approx 9.7 \times 10^2$ s ≈ 16 min.

QUESTION 139 [M] What is overvoltage in electrolysis?

ANSWER 139

Overvoltage (overpotential) is the extra potential beyond the reversible thermodynamic electrode potential needed to sustain electrolysis at a stated current density, due to kinetic and transport effects.

QUESTION 140 [H] Explain why electrolysis of aqueous sodium sulfate can produce gases rather than sodium metal.

ANSWER 140

In aqueous Na_2SO_4, water is reduced at the cathode to H_2 and oxidised at the anode to O_2; Na^+ and SO_4^{2-} mainly maintain conductivity. Water reduction is favoured over $Na^+ \rightarrow Na$ because sodium metal would immediately react with water.

QUESTION 141 [M] Why is lead dioxide used as the positive active material in a lead accumulator?

ANSWER 141

PbO_2 is the positive plate's oxidising active material: it accepts electrons during discharge and is reduced to $PbSO_4$, giving the cell a useful positive electrode potential.

QUESTION 142 [H] Compare the energy conversion in a hydrogen fuel cell and an electrolytic cell.

ANSWER 142

A hydrogen fuel cell spontaneously converts chemical energy of continuously supplied H_2/O_2 to electricity and water. An electrolytic cell consumes electrical energy to drive a non-spontaneous chemical change.

QUESTION 143 [M] Why is the standard electrode potential an intensive property?

ANSWER 143

Potential is energy per unit charge ($J\ C^{-1}$), independent of the total amount of electrode material or reaction; it is an intensive quantity.

QUESTION 144 [H] A concentration cell is formed using the same metal electrode in 0.01 M and 1.0 M solutions of its ion at 298 K. Explain the origin and direction of the cell potential.

ANSWER 144 · NEEDS QUALIFICATION

Different ion activities create a chemical-potential gradient. The dilute side is the anode (metal oxidises, raising its ion concentration); the concentrated side is the cathode (ions reduce and plate). Electrons flow dilute→concentrated externally. For M^{z+}, $E=(RT/zF)\ln(a_{high}/a_{low})$; no unique numeric value is possible without z and activity data.

UNIT 3: CHEMICAL KINETICS

Questions 145–216 · 72 complete pairs

QUESTION 145 [M] Define the rate of a chemical reaction.

ANSWER 145

Reaction rate is change in concentration of a reactant or product per unit time, with stoichiometric normalization and a negative sign for reactants so the rate is positive.

QUESTION 146 [M] Distinguish between average rate and instantaneous rate.

ANSWER 146

Average rate is concentration change divided by a finite time interval. Instantaneous rate is the derivative d(concentration)/dt at a particular time (with stoichiometric sign/normalization).

QUESTION 147 [H] Explain how the rate of disappearance of a reactant is related to the rate of formation of a product using stoichiometric coefficients.

ANSWER 147

For aA→bB, rate=−(1/a)d[A]/dt=(1/b)d[B]/dt; stoichiometric coefficients make the rates calculated from each species equal.

QUESTION 148 [M] Define rate law and rate constant.

ANSWER 148

Rate law is an experimentally determined relation such as rate=k[A]^m[B]^n. The rate constant k is the proportionality coefficient at a specified temperature; its units depend on overall order.

QUESTION 149 [M] What is the order of a reaction?

ANSWER 149

Reaction order is the sum of the experimentally measured exponents in the rate law; individual orders are the exponents of the corresponding concentrations.

QUESTION 150 [M] Define molecularity and distinguish it from order.

ANSWER 150

Molecularity is the number of reacting entities in a single elementary step and is a positive integer. Order is the sum of empirical rate-law exponents and applies to an overall reaction or step; it may be zero or fractional.

QUESTION 151 [H] Why can the order of a reaction be zero or fractional, whereas molecularity cannot?

ANSWER 151

Order is inferred from measured rate dependence and can be any real value, including zero or negative. Molecularity counts discrete species colliding in one elementary event, so it is a positive integer.

QUESTION 152 [M] What is a pseudo-first-order reaction? Give an example.

ANSWER 152

A pseudo-first-order reaction is actually of higher order but behaves first order when one reactant is in large excess and effectively constant; e.g. ester hydrolysis in excess water.

QUESTION 153 [H] Explain why hydrolysis of an ester in excess water can behave as a pseudo-first-order reaction.

ANSWER 153

Water is present in huge excess, so its concentration changes negligibly. In rate=k[ester][H_2O], absorb the nearly constant [H_2O] into k_{obs}=k[H_2O], giving rate=k_{obs}[ester].

QUESTION 154 [M] Write the integrated rate equation for a zero-order reaction.

ANSWER 154

For A→products at zero order: $[A]_t=[A]_0-kt$ (valid until reactant is depleted).

QUESTION 155 [M] Write the integrated rate equation for a first-order reaction.

ANSWER 155

For A→products at first order: $\ln([A]_0/[A]_t)=kt$, or $[A]_t=[A]_0e^{(-kt)}$.

QUESTION 156 [M] Write the half-life expression for a zero-order reaction.

ANSWER 156

For a zero-order reaction, $t_{1/2}=[A]_0/(2k)$.

QUESTION 157 [M] Write the half-life expression for a first-order reaction.

ANSWER 157

For a first-order reaction, $t_{1/2} = \ln 2/k = 0.693/k$.

QUESTION 158 [H] Show that the half-life of a first-order reaction is independent of the initial concentration.

ANSWER 158

At half-life, $[A]_t = [A]_0/2$. Substitute into $\ln([A]_0/[A]_t) = kt$: $\ln 2 = kt_{1/2}$, so $t_{1/2} = 0.693/k$, independent of $[A]_0$.

QUESTION 159 [H] A first-order reaction is 75% complete in 40 minutes. Calculate its half- life.

ANSWER 159 · NEEDS QUALIFICATION

75% complete means 25% remains, equal to one quarter of the initial amount: two half-lives have elapsed. Thus $t_{1/2} = 40/2 = 20$ min. Because "40 min" has ambiguous trailing-zero precision, 2×10^1 min is the 1-significant-figure form; 20 min assumes 2 significant figures.

QUESTION 160 [H] For a first-order reaction, 20% of the reactant remains after 30 minutes. Calculate the rate constant.

ANSWER 160 · NEEDS QUALIFICATION

$k = (1/t)\ln([A]_0/[A]_t) = \ln(1/0.20)/30 = \ln 5/30 = 0.05365\ \text{min}^{-1}$ unrounded; approximately $0.054\ \text{min}^{-1}$ if "30 min" is 2 significant figures, or $0.05\ \text{min}^{-1}$ if it is 1 significant figure (trailing-zero precision is unspecified).

QUESTION 161 [H] A zero-order reaction has $k = 2.0 \times 10^{-3}$ mol L^{-1} s^{-1} and initial concentration 0.10 mol L^{-1}. Calculate the time required for the concentration to fall to 0.04 mol L^{-1}.

ANSWER 161

For zero order, $[A]_t = [A]_0 - kt$. $t = (0.10 - 0.04)/(2.0 \times 10^{-3}) = 30$ s.

QUESTION 162 [M] What is the unit of the rate constant for a zero-order reaction?

ANSWER 162

Zero-order k has concentration/time units, e.g. mol L^{-1} s^{-1} (M s^{-1}).

QUESTION 163 [M] What is the unit of the rate constant for a first-order reaction?

ANSWER 163

First-order k has units of inverse time, e.g. s^{-1} or min^{-1}.

QUESTION 164 [H] Explain how a plot of concentration versus time can be used to identify a zero-order reaction.

ANSWER 164

A straight-line plot of [A] versus t identifies zero-order behaviour; its slope is −k and its intercept is $[A]_0$.

QUESTION 165 [H] Explain how a plot of ln[A] versus time can be used to identify a first- order reaction.

ANSWER 165

A straight-line plot of ln[A] versus t identifies first-order behaviour; its slope is −k and its intercept is $\ln[A]_0$.

QUESTION 166 [M] State the Arrhenius equation.

ANSWER 166

Arrhenius equation: $k=Ae^{(-E_a/RT)}$, or $\ln k=\ln A-E_a/(RT)$, where A is the frequency factor.

QUESTION 167 [M] Define activation energy.

ANSWER 167

Activation energy E_a is the energy barrier between reactants and the activated state along a reaction pathway; it is the minimum additional energy required in the simple Arrhenius picture.

QUESTION 168 [H] Explain how a catalyst changes the rate of a reaction without changing the equilibrium constant.

ANSWER 168

A catalyst provides a pathway with lower activation barriers for both forward and reverse reactions. It speeds both directions but leaves ΔG° and K unchanged, so equilibrium composition is unchanged.

QUESTION 169 [M] What is collision theory?

ANSWER 169

Collision theory models reaction as molecular collisions; only collisions with sufficient energy and suitable orientation are effective and form products.

QUESTION 170 [H] According to collision theory, why do not all molecular collisions lead to products?

ANSWER 170

Most collisions lack energy at least equal to the threshold/activation energy or have unsuitable orientation for bond rearrangement, so they do not form products.

QUESTION 171 [M] Define effective collision.

ANSWER 171

An effective collision has sufficient energy to overcome the activation barrier and an orientation that permits the required bonds to break/form.

QUESTION 172 [H] Explain the role of proper orientation in collision theory.

ANSWER 172

Even energetic molecules may not react if reactive sites do not meet in the proper geometry. Correct orientation is represented by a steric/orientation factor in collision theory.

QUESTION 173 [M] How does increasing concentration generally affect reaction rate?

ANSWER 173

Increasing reactant concentration generally increases collision frequency and, for positive-order species, raises the rate; the exact dependence is set by the rate law.

QUESTION 174 [M] How does increasing temperature generally affect reaction rate?

ANSWER 174

Increasing temperature generally raises the rate constant because a larger fraction of molecules has energy at or above E_a; rates usually increase substantially.

QUESTION 175 [H] Explain why a small rise in temperature can cause a large increase in rate constant.

ANSWER 175

The Arrhenius factor $e^{(-E_a/RT)}$ changes exponentially with $1/T$; heating appreciably increases the fraction of molecules able to cross the barrier, especially for large E_a.

QUESTION 176 [H] A reaction rate constant doubles when temperature increases from 300 K to 310 K. Explain how activation energy can be estimated from this information.

ANSWER 176

Use $\ln(k_2/k_1)=E_a/R(1/T_1-1/T_2)$. $E_a=8.314\ \ln 2/(1/300-1/310)=5.36\times10^4$ J $mol^{-1}\approx53.6$ kJ mol^{-1} (about 54 kJ mol^{-1} to 2 significant figures).

QUESTION 177 [M] What is the frequency factor in the Arrhenius equation?

ANSWER 177

A is the Arrhenius frequency factor (pre-exponential factor), reflecting collision frequency and orientation probability; it has the same units as k.

QUESTION 178 [H] Derive the two-temperature form of the Arrhenius equation.

ANSWER 178

From $\ln k=\ln A-E_a/(RT)$, subtract values at T_1 and T_2: $\ln(k_2/k_1)=-E_a/R(1/T_2-1/T_1)=E_a/R(1/T_1-1/T_2)$.

QUESTION 179 [M] What is the significance of a plot of ln k against 1/T?

ANSWER 179

For $\ln k$ versus $1/T$, slope$=-E_a/R$ and intercept$=\ln A$. The plot gives activation energy and frequency factor.

QUESTION 180 [H] How can activation energy be calculated from the slope of an Arrhenius plot?

ANSWER 180

Find slope m of $\ln k$ versus $1/T$; $E_a=-mR$. If $\log_{10}k$ is plotted instead, $E_a=-2.303Rm$.

QUESTION 181 [M] Distinguish between a slow reaction and a reaction with high activation energy.

ANSWER 181

A slow rate is an observed outcome that can also reflect low concentration or an unfavourable mechanism/pre-exponential factor. High E_a is a property of a pathway's energy barrier and often causes low rates at a given T, but the terms are not identical.

QUESTION 182 [H] Explain why the rate law must generally be determined experimentally.

ANSWER 182

An overall stoichiometric equation does not reveal the elementary steps or intermediates. Therefore rate exponents cannot generally be inferred from its coefficients and must be measured experimentally (except for an elementary step).

QUESTION 183 [M] For a reaction with rate = $k[A]^2[B]$, state the overall order.

ANSWER 183

Overall order=2+1=3 (second order in A and first order in B).

QUESTION 184 [H] For rate = $k[A]^2[B]$, predict the factor by which the rate changes if [A] is doubled and [B] is tripled.

ANSWER 184

Rate factor=$(2^2)(3^1)=12$; doubling [A] and tripling [B] makes the rate twelve times larger.

QUESTION 185 [M] What is the rate-determining step in a multistep reaction?

ANSWER 185

The rate-determining step is the slow step that constrains the overall reaction rate under the stated conditions; its position and preceding equilibria determine the observed law.

QUESTION 186 [H] Explain why the molecularity of an elementary step can be used to write its rate law, but the overall reaction equation usually cannot.

ANSWER 186

An elementary step is a single molecular event, so its molecularity specifies the colliding species and its rate law. An overall equation combines multiple steps and its coefficients need not be the kinetic exponents.

QUESTION 187 [M] What is the difference between a reaction mechanism and the overall reaction?

ANSWER 187

The overall reaction is the net stoichiometric change. A mechanism is the sequence of elementary steps, intermediates and transition states that produces that net change.

QUESTION 188 [H] A first-order reaction has $k = 6.93 \times 10^{-3}\ \text{min}^{-1}$. Calculate its half-life.

ANSWER 188

$t_{1/2} = 0.693/k = 0.693/(6.93\times10^{-3}\ \text{min}^{-1}) = 1.00\times10^{2}\ \text{min}$ (3 significant figures).

QUESTION 189 [M] What fraction of a first-order reactant remains after two half-lives?

ANSWER 189

After two half-lives, fraction remaining $= (1/2)^2 = 1/4 = 25\%$.

QUESTION 190 [H] A zero-order reaction takes 20 minutes for concentration to decrease from 0.10 M to 0.06 M. Calculate its rate constant.

ANSWER 190 · CORRECTED

For zero order, $k = ([A]_0 - [A]_t)/t = (0.10 - 0.06)/20 = 2\times10^{-3}\ \text{M min}^{-1}$ (to the precision of the 0.04 M concentration change; "20 min" is also precision-ambiguous).

QUESTION 191 [M] How does a catalyst affect the activation energy of the forward and reverse reactions?

ANSWER 191

A catalyst lowers the activation energy for both forward and reverse reactions through an alternative pathway; the change in barrier is consistent with the unchanged reaction ΔG.

QUESTION 192 [H] Explain why a catalyst does not alter the position of chemical equilibrium.

ANSWER 192

A catalyst accelerates forward and reverse reactions and helps equilibrium be reached sooner; it does not change ΔG° or K, so it cannot shift equilibrium composition.

QUESTION 193 [M] What is the difference between thermal and photochemical reactions in terms of energy input?

ANSWER 193

Thermal reactions receive energy through heat, usually a thermal molecular energy distribution. Photochemical reactions are initiated by absorption of photons, often producing electronically excited species.

QUESTION 194 [H] Explain the limitations of the elementary collision theory model for complex reactions.

ANSWER 194

Elementary collision theory treats molecules as simple colliding particles and is approximate for complex polyatomic molecules, solvent effects, energy redistribution, multi-step pathways and detailed orientation/entropy effects.

QUESTION 195 [M] For rate = $k[A][B]^2$, state the order with respect to each reactant and the overall order.

ANSWER 195

Order in A=1; order in B=2; overall order=1+2=3.

QUESTION 196 [H] If rate = $k[A]^2[B]$, what happens to the rate when [A] is halved and [B] is doubled?

ANSWER 196

Rate factor=$(1/2)^2\times2=1/2$. The rate becomes half its original value.

QUESTION 197 [M] Why is the rate constant independent of reactant concentration at a fixed temperature?

ANSWER 197

At a fixed temperature and for unchanged mechanism, k is the proportionality constant set by the pathway and temperature; concentration dependence is represented separately by concentration powers in the rate law.

QUESTION 198 [H] A first-order reaction is 90% complete in 50 min. Calculate k.

ANSWER 198 · NEEDS QUALIFICATION

90% complete leaves 10%. $k=\ln(10)/50=0.04605\ \text{min}^{-1}$ unrounded; approximately $0.046\ \text{min}^{-1}$ if "50 min" is 2 significant figures, or $0.05\ \text{min}^{-1}$ if it is 1 significant figure (trailing-zero precision is unspecified).

QUESTION 199 [M] How much of a first-order reactant remains after three half-lives?

ANSWER 199

After three half-lives, fraction remaining $=(1/2)^3=1/8=12.5\%$.

QUESTION 200 [H] A zero-order reaction has initial concentration 0.80 M and k = 0.02 M min^{-1}. Find its half-life.

ANSWER 200

For zero order, $t_{1/2}=[A]_0/(2k)=0.80/(2\times0.02)=20$ min.

QUESTION 201 [M] What does a negative slope in a first-order integrated-rate plot represent?

ANSWER 201

In a plot of ln[A] versus time, slope $=-k$. The negative sign reflects the decrease in reactant concentration; its magnitude is the first-order rate constant.

QUESTION 202 [H] Explain how half-life data at different initial concentrations can distinguish zero-order and first-order reactions.

ANSWER 202

For zero order, $t_{1/2}=[A]_0/(2k)$, so it increases with initial concentration. For first order, $t_{1/2}=0.693/k$ and is independent of initial concentration.

QUESTION 203 [M] Why does a catalyst provide an alternative reaction pathway?

ANSWER 203

A catalyst creates an alternative pathway with a lower activation barrier (or different sequence of steps), increasing rate without changing the net reaction equilibrium.

QUESTION 204 [H] Using the Arrhenius equation, explain how a plot of log k against 1/T differs in slope from a plot of ln k against 1/T.

ANSWER 204

For ln k versus 1/T, slope=$-E_a/R$. For $\log_{10}k$ versus 1/T, slope=$-E_a/(2.303R)$, smaller in magnitude by factor 2.303.

QUESTION 205 [M] What is the threshold energy in collision theory?

ANSWER 205

Threshold energy is the minimum collision energy required to reach a reactive state and form products; in the simple model it is associated with the activation barrier.

QUESTION 206 [H] Explain why a reaction with a large activation energy is especially sensitive to temperature.

ANSWER 206

The Boltzmann fraction of molecules with energy at least E_a depends exponentially on $-E_a/RT$. A larger E_a makes that fraction especially responsive to a temperature increase.

QUESTION 207 [M] Can a reaction have a negative order with respect to a reactant? Explain briefly.

ANSWER 207

Yes. A negative order can occur when a reactant inhibits the reaction, blocks catalyst sites, or changes a pre-equilibrium; it is an empirical rate-law effect, not a negative molecularity.

QUESTION 208 [H] Explain how product inhibition or adsorption can lead to unusual experimentally observed reaction orders.

ANSWER 208

Product binding can inhibit active sites, while adsorption can saturate or block a surface. Rates can then rise less than proportionally or even fall as concentration increases, producing zero or negative apparent orders.

QUESTION 209 [M] What is the difference between rate and rate constant?

ANSWER 209

Rate is the concentration change per time and usually varies as concentrations change. k is the rate-law proportionality constant at fixed temperature and mechanism, independent of reactant concentrations.

QUESTION 210 [H] For a reaction that is first order in A and zero order in B, write the rate law and describe the effect of doubling B.

ANSWER 210

rate=k[A]. Doubling B has no effect because B has zero order in the stated rate law.

QUESTION 211 [M] Why does the unit of the rate constant depend on reaction order?

ANSWER 211

Rate always has concentration/time units. Since rate=k(concentration)^n, k must have units concentration^(1−n) time^{-1} so the equation is dimensionally consistent.

QUESTION 212 [H] A first-order reaction has k = 0.0231 min^{-1}. Calculate the time for 90% completion.

ANSWER 212

90% complete leaves 10%; t=ln(10)/k=2.3026/0.0231=99.7 min (about 100 min).

QUESTION 213 [M] What is meant by the effective collision energy?

ANSWER 213 · CORRECTED

An effective collision has relative kinetic energy at least equal to the activation-energy threshold and suitable orientation. The energy available to cross the barrier is along the reaction coordinate; total kinetic energy alone does not ensure reaction.

QUESTION 214 [H] Explain the difference between activation energy and enthalpy change of a reaction.

ANSWER 214

E_a is the forward/reverse barrier from a reactant state to the transition state and is non-negative for a pathway. ΔH is the energy difference between products and reactants and may be positive or negative; they are not generally equal.

QUESTION 215 [M] Why can catalysts be selective?

ANSWER 215

Catalysts can bind selected reactants or transition states and provide a lower-energy pathway only for particular structures, so they may accelerate some reactions much more than others.

QUESTION 216 [H] For a first-order reaction, the time required for 99% completion is how many times its half-life? Derive the result.

ANSWER 216

For first order, 99% completion leaves $0.01[A]_0$, so $t_{99}=\ln 100/k$. Since $t_{1/2}=\ln 2/k$, $t_{99}/t_{1/2}=\ln 100/\ln 2=6.64$ half-lives.

UNIT 4: d- AND f-BLOCK ELEMENTS

Questions 217–288 · 72 complete pairs

QUESTION 217 [M] Define transition elements in terms of partially filled d orbitals.

ANSWER 217

A transition element is an element whose atom has an incomplete d subshell or which forms at least one common cation with an incomplete d subshell.

QUESTION 218 [M] Write the general electronic configuration of d-block elements.

ANSWER 218

General d-block configuration: $(n-1)d^{1-10}\ ns^{0-2}$; exceptions occur because the d and s subshell energies are close.

QUESTION 219 [H] Why are Zn, Cd and Hg generally not considered transition elements under the strict definition?

ANSWER 219

Zn, Cd and Hg atoms and their common M^{2+} ions have filled d^{10} subshells, not partially filled d subshells; they therefore fail the strict transition-element definition.

QUESTION 220 [M] Why do transition metals show variable oxidation states?

ANSWER 220

The ns and (n−1)d orbital energies are close, so different numbers of these electrons can participate in bonding or be removed, giving multiple oxidation states.

QUESTION 221 [H] Explain why the +2 oxidation state is common among first-row transition metals.

ANSWER 221

The two outer ns electrons are usually lost first, giving M^{2+}. The (n−1)d electrons are more tightly held, so +2 is common for first-row metals.

QUESTION 222 [M] Why are many transition-metal ions coloured?

ANSWER 222

Partly filled d subshells have split d-orbital energy levels in a ligand field. Absorption of visible light promotes d–d transitions; remaining transmitted/reflected light gives colour.

QUESTION 223 [H] Explain d–d transitions and their role in the colour of transition-metal complexes.

ANSWER 223

Ligand fields split the metal d orbitals. A photon with energy matching the splitting promotes an electron between them; selective visible absorption determines observed colour.

QUESTION 224 [M] Why are d^0 and d^{10} ions often colourless?

ANSWER 224

A d^0 ion has no d electron to excite and a d^{10} ion has no vacant d state within the split d set; thus d–d transitions are unavailable. Charge-transfer bands can still colour some compounds.

QUESTION 225 [H] Explain why the colour of a transition-metal ion can change with ligand or oxidation state.

ANSWER 225

Changing oxidation state changes d-electron count and splitting; changing ligand alters the ligand-field strength and Δ. Both change transition energies and the wavelengths absorbed.

QUESTION 226 [M] Why do transition metals often act as catalysts?

ANSWER 226

Transition metals can access multiple oxidation states, adsorb reactants on surfaces and form intermediates, providing alternative pathways with lower activation barriers.

QUESTION 227 [H] Explain how variable oxidation states and surface adsorption contribute to catalytic activity.

ANSWER 227

Variable oxidation states allow electron transfer between reactants and metal centres. Adsorption brings reactants together and weakens bonds at the surface, lowering activation barriers.

QUESTION 228 [M] What is the origin of paramagnetism in transition-metal ions?

ANSWER 228

Unpaired d electrons have magnetic moments; their alignment with an applied field causes paramagnetism. Ions with all electrons paired are diamagnetic.

QUESTION 229 [H] Calculate the spin-only magnetic moment for an ion containing three unpaired electrons.

ANSWER 229

Spin-only moment $\mu=\sqrt{[n(n+2)]}$ BM. For $n=3$, $\mu=\sqrt{15}=3.87$ BM.

QUESTION 230 [M] Why do transition metals form alloys readily?

ANSWER 230

Many transition metals have similar atomic sizes and crystal structures, so atoms can replace one another in metallic lattices to form substitutional alloys.

QUESTION 231 [M] What are interstitial compounds? Give two characteristic properties.

ANSWER 231

Interstitial compounds form when small atoms such as H, C or N occupy holes in a metal lattice. They are often hard/high-melting and may be non-stoichiometric.

QUESTION 232 [H] Explain why interstitial compounds are often hard and have high melting points.

ANSWER 232

Small atoms fit into metal-lattice interstices and strongly distort/strengthen the lattice, restricting metal-atom motion; this often gives high hardness and melting points.

QUESTION 233 [M] Describe the general trend in atomic radii across the first transition series.

ANSWER 233

Across the first transition series, atomic radii decrease modestly at first, vary little through the middle, and may rise slightly near the end; added d electrons partly shield increasing nuclear charge.

QUESTION 234 [H] Why is the variation in ionisation enthalpy across the first transition series irregular?

ANSWER 234

Increasing nuclear charge, imperfect d-electron shielding, electron–electron repulsion and irregular stability of d configurations compete, producing a non-smooth ionisation-enthalpy trend.

QUESTION 235 [M] What is lanthanoid contraction?

ANSWER 235

Lanthanoid contraction is the gradual decrease in atomic and especially Ln^{3+} ionic radii from La to Lu as nuclear charge increases across the series.

QUESTION 236 [H] Explain the cause of lanthanoid contraction in terms of shielding by 4f electrons.

ANSWER 236

Added 4f electrons shield the increasing nuclear charge poorly. Effective nuclear charge therefore rises across the series and contracts the electron cloud.

QUESTION 237 [M] State two consequences of lanthanoid contraction.

ANSWER 237

Lanthanoid contraction makes Zr and Hf nearly equal in size, so they have very similar chemistry and are difficult to separate; it also affects hydroxide basicity and other size-dependent properties.

QUESTION 238 [H] Explain the similarity in radii of zirconium and hafnium.

ANSWER 238

The 4f electrons inserted between La and Lu shield poorly, causing lanthanoid contraction. Consequently Hf, after the lanthanoids, has almost the same radius as Zr above it.

QUESTION 239 [M] Write the general electronic configuration of lanthanoids.

ANSWER 239

General lanthanoid configuration: $[Xe]4f^{0-14}5d^{0-1}6s^{2}$; most Ln atoms have 4f electrons, with limited 5d participation.

QUESTION 240 [M] Why is +3 the most common oxidation state of lanthanoids?

ANSWER 240

Lanthanoids readily lose two 6s electrons and one 4f/5d electron to form Ln^{3+}. This oxidation state is common because the resulting ions are generally stable in compounds.

QUESTION 241 [H] Explain why some lanthanoids also exhibit +2 or +4 oxidation states.

ANSWER 241

+2 or +4 can provide especially stable empty, half-filled or filled 4f subshells, e.g. Ce^{4+} ($4f^0$), Eu^{2+} ($4f^7$), Tb^{4+} ($4f^7$), Yb^{2+} ($4f^{14}$); their stability also depends on lattice/hydration energies.

QUESTION 242 [M] Write the general electronic configuration of actinoids.

ANSWER 242

General actinoid configuration: $[Rn]5f^{0-14}6d^{0-2}7s^2$, with substantial variation among elements.

QUESTION 243 [H] Why do actinoids show a wider range of oxidation states than lanthanoids?

ANSWER 243

Actinoid 5f, 6d and 7s orbitals are closer in energy and 5f electrons are more available for bonding than 4f electrons in lanthanoids; several electron counts can therefore participate.

QUESTION 244 [M] Give two differences between lanthanoids and actinoids.

ANSWER 244

Actinoids fill 5f orbitals whereas lanthanoids fill 4f; actinoids show a wider range of oxidation states and are all radioactive, while many lanthanoids have stable isotopes and predominantly +3 chemistry.

QUESTION 245 [H] Explain why most actinoids are radioactive.

ANSWER 245

Actinoids have very heavy nuclei with large proton–proton repulsion; their nuclei are unstable and undergo radioactive decay toward more stable configurations. All actinoids are radioactive, though half-lives vary greatly.

QUESTION 246 [M] What is the colour of potassium dichromate solution and which chromium oxidation state is present?

ANSWER 246

Aqueous potassium dichromate is orange; chromium is in oxidation state +6.

QUESTION 247 [M] What is the colour of permanganate ion in aqueous solution?

ANSWER 247

Permanganate ion MnO_4^- is deep purple/violet in aqueous solution.

QUESTION 248 [H] Write the balanced ionic equation for the reduction of dichromate ion to Cr^{3+} in acidic medium.

ANSWER 248

$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$.

QUESTION 249 [H] Write the balanced ionic equation for the reduction of permanganate ion to Mn^{2+} in acidic medium.

ANSWER 249

$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$.

QUESTION 250 [M] Describe the oxidising action of acidified potassium permanganate.

ANSWER 250

Acidified $KMnO_4$ accepts electrons and oxidises many reductants (e.g. $Fe^{2+} \rightarrow Fe^{3+}$ or $I^- \rightarrow I_2$); MnO_4^- is reduced mainly to pale Mn^{2+} in sufficiently acidic solution.

QUESTION 251 [H] Explain why the products of permanganate reduction depend on the pH of the medium.

ANSWER 251

H^+ participates in the reduction half-reaction, so its concentration shifts the redox equilibrium and changes the reduction potential/products. Acid favours Mn^{2+}; neutral/weakly alkaline conditions commonly yield MnO_2; strongly alkaline conditions can yield MnO_4^{2-}.

QUESTION 252 [M] State the oxidation state of chromium in $K_2Cr_2O_7$.

ANSWER 252

In $K_2Cr_2O_7$: $2(+1)+2x+7(-2)=0$, so $x=+6$ for chromium.

QUESTION 253 [M] State the oxidation state of manganese in $KMnO_4$.

ANSWER 253

In $KMnO_4$: $(+1)+x+4(-2)=0$, so manganese is +7.

QUESTION 254 [H] Calculate the oxidation state of manganese in MnO_4^{2-} and compare it with MnO_4^-.

ANSWER 254

For MnO_4^{2-}, $x+4(-2)=-2$, so Mn=+6. In MnO_4^-, Mn=+7; manganate is one oxidation state lower.

QUESTION 255 [M] Why is Cr^{2+} a reducing agent while Mn^{3+} can act as an oxidising agent?

ANSWER 255

Cr^{2+} (d^4) readily loses an electron to form particularly stable Cr^{3+} (d^3), so it reduces other species. Mn^{3+} (d^4) readily gains an electron to form half-filled Mn^{2+} (d^5), so it can oxidise other species.

QUESTION 256 [H] Use electronic configurations to explain the relative stability of Cr^{3+} and Mn^{2+}.

ANSWER 256

Cr^{3+} is d^3, with a stable t_{2g}^3 arrangement in an octahedral field; Mn^{2+} is d^5 and gains half-filled-subshell stability. Their stability also depends on environment and hydration.

QUESTION 257 [M] Why do transition metals have high enthalpies of atomisation?

ANSWER 257

Several d and s electrons can contribute to metal–metal bonding; strong cohesive bonding requires substantial energy to separate atoms, giving high atomisation enthalpies.

QUESTION 258 [H] Explain the trend in melting points across the first-row transition series and why it is not perfectly regular.

ANSWER 258

Melting points generally rise toward the middle of the first transition series, where metallic bonding is strong, then fall toward the end as d participation/cohesion declines. Subshell configurations, structure and bonding make the trend irregular.

QUESTION 259 [M] What is disproportionation? Give a transition-metal example.

ANSWER 259

Disproportionation is simultaneous oxidation and reduction of the same oxidation state, e.g. $2Cu^{+}(aq) \rightarrow Cu^{2+}(aq) + Cu(s)$.

QUESTION 260 [H] Explain why Cu^{+} tends to disproportionate in aqueous solution under standard conditions.

ANSWER 260

In water, Cu^{2+} is strongly stabilised by hydration relative to Cu^{+}; the free-energy balance makes $2Cu^{+} \rightarrow Cu^{2+} + Cu$ favourable under standard aqueous conditions.

QUESTION 261 [M] Why are transition-metal compounds frequently used as pigments?

ANSWER 261

Many transition-metal ions have partly filled d levels with ligand-field or charge-transfer absorptions in visible light; selective absorption produces vivid colours useful in pigments.

QUESTION 262 [H] Compare the magnetic behaviour expected for high-spin d^5 and low-spin d^6 complexes.

ANSWER 262

High-spin d^5 has five unpaired electrons, $\mu=\sqrt{35}=5.92$ BM, and is strongly paramagnetic. Low-spin d^6 has t_{2g}^6, no unpaired electrons, $\mu\approx0$, and is diamagnetic (octahedral spin-only model).

QUESTION 263 [M] What is the difference between lanthanoid and actinoid contraction?

ANSWER 263

Lanthanoid contraction is due to increasing nuclear charge poorly shielded by 4f electrons; actinoid contraction arises similarly from poor 5f shielding. Both shrink trivalent ionic radii across their series; actinoid behaviour is less regular because 5f participation is greater.

QUESTION 264 [H] Explain why separation of lanthanoids is difficult and how contraction helps distinguish their properties.

ANSWER 264

Ln^{3+} ions have similar charge and only gradually changing radii/chemistry, making their separation difficult. The small systematic contraction does produce gradual differences in basicity, complexation and related properties that can be exploited.

QUESTION 265 [M] Why is the +6 oxidation state accessible to chromium in chromate and dichromate ions?

ANSWER 265

Chromium has 3d and 4s valence electrons available, and strongly electronegative oxygen stabilises high oxidation state +6 in chromate/dichromate oxyanions.

QUESTION 266 [H] Explain the interconversion of chromate and dichromate ions with change in pH.

ANSWER 266

$2CrO_4^{2-}+2H^+\rightleftharpoons Cr_2O_7^{2-}+H_2O$. Acid shifts yellow chromate toward orange dichromate; added base removes H^+ and shifts back toward chromate.

QUESTION 267 [M] Write the electronic configuration of Cr and explain its exceptional configuration.

ANSWER 267

Cr: $[Ar]3d^5 4s^1$ rather than $3d^4 4s^2$. The 3d and 4s energies are close, and the half-filled d^5 subshell provides extra stability.

QUESTION 268 [H] Explain the exceptional electronic configurations of Cr and Cu in terms of subshell stability.

ANSWER 268

Cr is $[Ar]3d^5 4s^1$ and Cu is $[Ar]3d^{10} 4s^1$ rather than the simple expected distributions; near-degenerate 4s/3d levels allow configurations with half-filled or filled d subshells.

QUESTION 269 [M] Why is the third ionisation enthalpy of manganese relatively high after formation of Mn^{2+}?

ANSWER 269

Mn^{2+} is $[Ar]3d^5$, a particularly stable half-filled configuration. Removing a third electron breaks that arrangement, so the third ionisation enthalpy is relatively high.

QUESTION 270 [H] Compare the stability of Fe^{2+} and Fe^{3+} using electronic configurations.

ANSWER 270

Fe^{2+} is d^6; Fe^{3+} is d^5, a half-filled subshell, which often favours Fe^{3+}. Relative stability is environment-dependent because hydration, lattice energy and ligand fields also contribute.

QUESTION 271 [M] What is the common oxidation state of scandium and why?

ANSWER 271

Sc commonly forms Sc^{3+} by losing its two 4s and one 3d electron, leaving the stable [Ar] configuration.

QUESTION 272 [H] Explain why the highest oxidation states are more common for early transition elements than for later ones.

ANSWER 272

Early transition metals have fewer d electrons and can use more valence electrons in bonding, making high oxidation states more accessible. Later metals have higher ionisation costs and increasingly stable lower d configurations.

QUESTION 273 [M] Why do transition-metal ions form complex ions readily?

ANSWER 273

Transition-metal ions are often small with high charge density and vacant/available orbitals; they can accept ligand lone pairs and form coordinate bonds.

QUESTION 274 [H] Explain the relationship between d-electron configuration and the magnetic behaviour of transition-metal ions.

ANSWER 274

The number and arrangement of unpaired d electrons determine spin-only moment $\mu = \sqrt{[n(n+2)]}$ BM. Any unpaired electrons give paramagnetism; all paired gives diamagnetism; ligand strength can change pairing in complexes.

QUESTION 275 [M] Why is Cu^{2+} coloured but Zn^{2+} colourless?

ANSWER 275

Cu^{2+} is d^9 with a partially filled d level, allowing visible d–d transitions. Zn^{2+} is d^{10} with no d–d transition and is usually colourless unless charge-transfer absorption occurs.

QUESTION 276 [H] Explain why d–d transition energies are affected by ligand identity.

ANSWER 276

Ligand identity sets ligand-field strength through its bonding and interaction with metal d orbitals, changing Δ. The absorbed photon energy and d–d transition wavelength therefore change.

QUESTION 277 [M] What is the effect of lanthanoid contraction on basicity of lanthanoid hydroxides?

ANSWER 277

Basicity of lanthanoid hydroxides generally decreases from $La(OH)_3$ to $Lu(OH)_3$ as Ln^{3+} radius contracts and polarising power/covalent character increase.

QUESTION 278 [H] Explain the trend in basic strength of lanthanoid hydroxides across the series.

ANSWER 278

Across La to Lu, Ln^{3+} size decreases while charge remains +3. Greater polarisation strengthens Ln–O covalent character and reduces hydroxide basicity; $La(OH)_3$ is more basic than $Lu(OH)_3$.

QUESTION 279 [M] Why are actinoids more prone to complex formation than lanthanoids?

ANSWER 279

Actinoid 5f orbitals are more spatially extended and can participate in bonding, and actinoids commonly access higher oxidation states and strong metal–ligand interactions; this favours complex formation.

QUESTION 280 [H] Explain why actinoid chemistry is more complicated due to 5f, 6d and 7s orbital energies.

ANSWER 280

The 5f, 6d and 7s levels are close in energy, so several configurations and oxidation states are accessible. This, plus greater 5f bonding participation and radioactivity, makes actinoid chemistry complex.

QUESTION 281 [M] What is the role of sulfuric acid in preparing potassium dichromate from chromite ore?

ANSWER 281

Chromite ore is oxidised/roasted to soluble chromate. H_2SO_4 acidifies the chromate solution, shifting $2CrO_4^{2-} + 2H^+ \rightleftharpoons Cr_2O_7^{2-} + H_2O$ toward dichromate; adding KCl then crystallises less-soluble $K_2Cr_2O_7$.

QUESTION 282 [H] Balance the redox reaction between MnO_4^- and Fe^{2+} in acidic medium.

ANSWER 282

$MnO_4^- + 5Fe^{2+} + 8H^+ \rightarrow Mn^{2+} + 5Fe^{3+} + 4H_2O$.

QUESTION 283 [M] Why is permanganate a stronger oxidising agent in acidic medium than in neutral medium?

ANSWER 283

In acid, H^+ drives the MnO_4^-/Mn^{2+} reduction half-reaction, whose potential is more positive; permanganate is consequently a stronger oxidant than when reduced to MnO_2 in neutral medium.

QUESTION 284 [H] Explain the use of $K_2Cr_2O_7$ as a primary standard in redox titrations.

ANSWER 284

$K_2Cr_2O_7$ is available in high purity, stable in air, non-hygroscopic, has a relatively high molar mass and can be weighed accurately to prepare a standard solution.

QUESTION 285 [M] Why are transition metals usually good conductors of heat and electricity?

ANSWER 285

Delocalised mobile electrons carry charge and transport thermal energy through the metallic lattice, giving transition metals generally good electrical and thermal conductivity.

QUESTION 286 [H] Explain why the enthalpies of atomisation of Zn and Cd are comparatively lower than those of many transition metals.

ANSWER 286

Zn and Cd have filled $(n-1)d^{10}$ subshells and fewer unpaired d electrons to strengthen metal–metal cohesion; their metallic bonding and atomisation enthalpies are comparatively lower.

QUESTION 287 [M] What is the difference between a transition metal and an inner-transition metal?

ANSWER 287

Transition metals involve filling d orbitals in the outer transition series; inner-transition metals involve progressive filling of inner f orbitals (lanthanoids 4f, actinoids 5f).

QUESTION 288 [H] Explain why the standard electrode potential of a transition-metal couple cannot be predicted from electronic configuration alone.

ANSWER 288

E° reflects several energetic terms—atomisation, ionisation, hydration/solvation and entropy—not just the electron configuration. Their balance depends on the species and medium, so configuration alone is insufficient.

UNIT 5: COORDINATION COMPOUNDS

Questions 289–358 · 70 complete pairs

QUESTION 289 [M] Define a coordination compound and give one example.

ANSWER 289

A coordination compound contains a central metal atom/ion bonded to surrounding ligands by coordinate bonds; e.g. $[Co(NH_3)_6]Cl_3$.

QUESTION 290 [M] What is a ligand? Classify ligands on the basis of denticity.

ANSWER 290

A ligand donates one or more lone pairs to a central metal. By denticity, ligands are monodentate (one donor atom), bidentate (two) or polydentate (several); e.g. NH_3, en and EDTA.

QUESTION 291 [M] Define coordination number with an example.

ANSWER 291

Coordination number is the number of ligand donor atoms directly bonded to the metal. In $[Co(NH_3)_6]^{3+}$ it is 6.

QUESTION 292 [H] Determine the oxidation state of the central metal in $[Co(NH_3)_5Cl]Cl_2$.

ANSWER 292

In $[Co(NH_3)_5Cl]Cl_2$ the complex cation has charge +2. With NH_3 neutral and coordinated Cl^- at −1, $x-1=+2$, so Co is +3.

QUESTION 293 [M] Distinguish between a double salt and a coordination compound.

ANSWER 293

A double salt dissociates in water into all constituent simple ions and loses its identity. A coordination compound retains complex ions in solution, e.g. $[Fe(CN)_6]^{4-}$.

QUESTION 294 [M] State Werner's primary and secondary valencies.

ANSWER 294

Werner's primary valency is ionisable and corresponds mainly to oxidation state; secondary valency is coordination number, satisfied by ligands and directional in space.

QUESTION 295 [H] Explain how Werner's theory accounts for the structures of $CoCl_3{\cdot}6NH_3$ and $CoCl_3{\cdot}5NH_3$.

ANSWER 295

$CoCl_3{\cdot}6NH_3$ is $[Co(NH_3)_6]Cl_3$; six secondary valencies and three ionisable Cl^-. $CoCl_3{\cdot}5NH_3$ is $[Co(NH_3)_5Cl]Cl_2$; five NH_3 plus one coordinated Cl^- give coordination number 6, with two ionisable Cl^-.

QUESTION 296 [M] Write the IUPAC name of $[Co(NH_3)_6]Cl_3$.

ANSWER 296

Hexaamminecobalt(III) chloride.

QUESTION 297 [M] Write the IUPAC name of $K_4[Fe(CN)_6]$.

ANSWER 297

Potassium hexacyanidoferrate(II).

QUESTION 298 [H] Write the formula of pentaamminechloridocobalt(III) chloride.

ANSWER 298

$[Co(NH_3)_5Cl]Cl_2$.

QUESTION 299 [M] What is a chelating ligand? Give an example.

ANSWER 299

A chelating ligand binds a metal through two or more donor atoms, forming one or more rings; ethane-1,2-diamine (en) or oxalate is an example.

QUESTION 300 [H] Explain why chelate complexes are often more stable than analogous complexes with monodentate ligands.

ANSWER 300

Multidentate ligands form several metal–donor bonds at once, and chelation often has a favourable entropy because several coordinated monodentate molecules are released as one multidentate ligand binds. Ring formation and rebinding make dissociation less favourable than for analogous monodentate ligands.

QUESTION 301 [M] What is the difference between homoleptic and heteroleptic complexes?

ANSWER 301

Homoleptic complexes contain only one kind of ligand; heteroleptic complexes contain two or more different ligand types.

QUESTION 302 [M] Explain the bonding in coordination compounds according to valence bond theory.

ANSWER 302

In VBT, ligands donate lone pairs into vacant hybrid orbitals on the metal. Hybrid-orbital arrangement predicts geometry; the metal's unpaired d electrons help predict magnetic behaviour.

QUESTION 303 [H] Predict the hybridisation and geometry of $[Ni(CN)_4]^{2-}$ using VBT.

ANSWER 303

In $[Ni(CN)_4]^{2-}$, Ni^{2+} is d^8 and CN^- is strong-field; d electrons pair, giving dsp^2 hybridisation, square-planar geometry and diamagnetism.

QUESTION 304 [H] Predict the hybridisation and magnetic behaviour of $[NiCl_4]^{2-}$ using VBT.

ANSWER 304

In $[NiCl_4]^{2-}$, Ni^{2+} is d^8 and Cl^- weak-field; the complex is sp^3 tetrahedral with two unpaired electrons and is paramagnetic.

QUESTION 305 [M] What is crystal field splitting?

ANSWER 305

Crystal-field splitting is removal of degeneracy of metal d orbitals by the electrostatic field of surrounding ligands; the energy separation depends on geometry and ligand strength.

QUESTION 306 [M] Distinguish between strong-field and weak-field ligands.

ANSWER 306

Strong-field ligands produce a large splitting Δ and may cause electron pairing; weak-field ligands give a smaller Δ, often leaving electrons unpaired in high-spin complexes.

QUESTION 307 [H] Explain why $[Fe(CN)_6]^{4-}$ is low spin while $[FeF_6]^{4-}$ is generally high spin.

ANSWER 307

Both contain Fe^{2+} (d^6). CN^- gives $\Delta_o > P$, so $[Fe(CN)_6]^{4-}$ is low-spin t_{2g}^6; weak-field F^- gives $\Delta_o < P$, so $[FeF_6]^{4-}$ is high-spin $t_{2g}^4 e_g^2$.

QUESTION 308 [M] Draw or describe the splitting of d orbitals in an octahedral field.

ANSWER 308

In an octahedral field, the five d orbitals split into lower-energy t_{2g} (d_{xy}, d_{xz}, d_{yz}) and higher-energy e_g ($d_{z^2}, d_{x^2-y^2}$), separated by Δ_o.

QUESTION 309 [M] Draw or describe the splitting of d orbitals in a tetrahedral field.

ANSWER 309

In a tetrahedral field, e ($d_{z^2}, d_{x^2-y^2}$) is lower and t_2 (d_{xy}, d_{xz}, d_{yz}) higher; the order is opposite to octahedral and the gap is Δ_t.

QUESTION 310 [H] Why is tetrahedral splitting smaller than octahedral splitting for the same metal and ligand?

ANSWER 310

In tetrahedral geometry only four ligands approach and none lies directly on the axes; interaction with d lobes is smaller. Consequently $\Delta_t \approx (4/9)\Delta_o$ for the same metal/ligand (approximately).

QUESTION 311 [M] What is crystal field stabilisation energy?

ANSWER 311

Crystal-field stabilisation energy is the net lowering in d-electron energy relative to the unsplit barycentre due to occupation of the lower-energy ligand-field orbitals.

QUESTION 312 [H] Calculate the CFSE for an octahedral d^3 ion, ignoring pairing energy.

ANSWER 312

Octahedral d^3 is $t_{2g}^3 e_g^0$. $CFSE = 3(-0.4\Delta_o) = -1.2\Delta_o$, i.e. stabilisation of $1.2\Delta_o$; pairing energy is excluded.

QUESTION 313 [M] Why are many coordination compounds coloured?

ANSWER 313

Partly filled d orbitals can undergo ligand-field d–d transitions by absorbing visible photons, and charge-transfer transitions may also absorb light; unabsorbed light gives observed colour.

QUESTION 314 [H] Explain how ligand-field splitting determines the wavelength of light absorbed by a complex.

ANSWER 314

A photon is absorbed when hc/λ matches the ligand-field transition energy (approximately Δ). Thus $\lambda = hc/\Delta$; larger splitting corresponds to shorter absorbed wavelength.

QUESTION 315 [M] How can magnetic measurements help distinguish high-spin and low- spin complexes?

ANSWER 315

Magnetic susceptibility/moment estimates unpaired-electron count. High-spin complexes usually have more unpaired electrons and larger paramagnetic moments than their low-spin counterparts; all-paired complexes are diamagnetic.

QUESTION 316 [H] Calculate the spin-only magnetic moment of a complex with four unpaired electrons.

ANSWER 316

Spin-only $\mu = \sqrt{[n(n+2)]}$ BM $= \sqrt{[4\times6]} = \sqrt{24} = 4.90$ BM.

QUESTION 317 [M] What is geometrical isomerism in coordination compounds?

ANSWER 317

Geometrical isomerism occurs when the same ligands occupy different relative positions around the metal, e.g. cis and trans arrangements, without changing connectivity.

QUESTION 318 [M] What is optical isomerism in coordination compounds?

ANSWER 318

Optical isomers are non-superimposable mirror-image coordination structures (enantiomers) that rotate plane-polarised light in opposite directions.

QUESTION 319 [H] Explain cis–trans isomerism in square-planar complexes of type $[MA_2B_2]$.

ANSWER 319

Square-planar $[MA_2B_2]$ has cis isomer with identical A ligands adjacent and trans isomer with A ligands opposite; these are distinct spatial arrangements.

QUESTION 320 [H] How many geometrical isomers are possible for an octahedral complex of type $[MA_4B_2]$?

ANSWER 320

Two: cis-$[MA_4B_2]$ and trans-$[MA_4B_2]$, according to the relative positions of the two B ligands.

QUESTION 321 [M] Give an example of a coordination compound showing linkage isomerism.

ANSWER 321

$[Co(NH_3)_5(NO_2)]Cl_2$ can bind NO_2^- through N (nitro) or O (nitrito), giving linkage isomers.

QUESTION 322 [M] What is ionisation isomerism?

ANSWER 322

Ionisation isomers exchange a ligand inside the coordination sphere with a counter-ion outside, producing different ions in solution; e.g. $[Co(NH_3)_5Br]SO_4$ and $[Co(NH_3)_5SO_4]Br$.

QUESTION 323 [H] Distinguish between linkage, ionisation and hydrate isomerism with examples.

ANSWER 323

Linkage: $[Co(NH_3)_5(NO_2)]Cl_2$ (nitro/nitrito donor atom). Ionisation: $[Co(NH_3)_5Br]SO_4$ vs $[Co(NH_3)_5SO_4]Br$. Hydrate: $[Cr(H_2O)_6]Cl_3$ vs $[Cr(H_2O)_5Cl]Cl_2{\cdot}H_2O$.

QUESTION 324 [M] Explain the role of coordination compounds in qualitative analysis.

ANSWER 324

Complex formation gives characteristic colours or selective precipitates and can mask interfering ions, allowing ions to be separated or identified in qualitative analysis.

QUESTION 325 [M] How is complex formation used in the extraction of gold or silver?

ANSWER 325

In cyanide leaching, Au/Ag forms soluble dicyano complexes (e.g. $4Au+8CN^-+O_2+2H_2O \rightarrow 4[Au(CN)_2]^-+4OH^-$); metal is then recovered from solution.

QUESTION 326 [M] State the role of haemoglobin as a coordination compound.

ANSWER 326

Haemoglobin's Fe-containing heme coordination sites bind O_2 reversibly, enabling oxygen transport in blood; binding must remain reversible and not permanently oxidise the iron.

QUESTION 327 [H] Explain the role of chlorophyll and vitamin B_{12} in biological systems in terms of their central metal ions.

ANSWER 327

Chlorophyll has central Mg^{2+} in a porphyrin-like coordination framework for light capture in photosynthesis. Vitamin B_{12} (cobalamin) contains Co coordinated within a corrin ring and is essential in several enzyme-mediated reactions.

QUESTION 328 [M] Why does EDTA form stable complexes with many metal ions?

ANSWER 328

EDTA is hexadentate and can bind through six donor atoms, forming multiple bonds and chelate rings around many metal ions; this yields stable complexes and strong sequestration.

QUESTION 329 [H] Compare inner-orbital and outer-orbital octahedral complexes in VBT.

ANSWER 329

Inner-orbital octahedral complexes use two $(n-1)d$, one ns and three np orbitals (d^2sp^3), often low spin. Outer-orbital complexes use ns, np and nd (sp^3d^2), often high spin; this is VBT's model.

QUESTION 330 [M] What is the spectrochemical series?

ANSWER 330

The spectrochemical series orders ligands by increasing ability to split metal d orbitals, e.g. $I^- < Br^- < Cl^- < F^- < OH^- < H_2O < NH_3 < en < CN^- < CO$.

QUESTION 331 [H] Explain why pairing may occur in a strong ligand field but not in a weak ligand field.

ANSWER 331

If $\Delta > P$, placing an electron in a higher orbital costs more than pairing in a lower one, so pairing occurs. If $\Delta < P$, occupation of separate higher orbitals costs less, favouring high spin.

QUESTION 332 [M] Determine the oxidation state and coordination number of Fe in $[Fe(CN)_6]^{3-}$.

ANSWER 332

For $[Fe(CN)_6]^{3-}$, $x+6(-1)=-3$ gives Fe=+3. Six monodentate CN^- donor atoms give coordination number 6.

QUESTION 333 [H] Predict the number of unpaired electrons in $[Mn(CN)_6]^{3-}$ using crystal field theory.

ANSWER 333

Mn in $[Mn(CN)_6]^{3-}$ is +3, d^4. Strong-field CN^- gives low-spin $t_2g^4e_g^0$, with two unpaired electrons.

QUESTION 334 [M] What is the effective atomic number rule? State one limitation.

ANSWER 334

EAN=metal atomic number−oxidation state+electrons donated by ligands; stability is sometimes associated with reaching a noble-gas electron count, but it is only an empirical guideline.

QUESTION 335 [H] Explain why the EAN rule does not reliably predict the stability of every coordination compound.

ANSWER 335

Many stable complexes do not attain noble-gas EAN; ligand field, covalency, steric effects, charge, kinetics and thermodynamics can dominate. The rule is not a universal stability criterion.

QUESTION 336 [M] Distinguish between a monodentate and a bidentate ligand.

ANSWER 336

A monodentate ligand binds through one donor atom; a bidentate ligand binds through two donor atoms to the same metal, often forming a chelate ring.

QUESTION 337 [H] Explain why $[Pt(NH_3)_2Cl_2]$ has cis and trans forms and state the relevance of the cis form.

ANSWER 337

Square-planar Pt(II) complexes of formula $[Pt(NH_3)_2Cl_2]$ have cis (like ligands adjacent) and trans (opposite) arrangements. Cisplatin is the cis form and is an important anticancer drug; the trans form is not equivalently effective.

QUESTION 338 [M] Find the oxidation state of Co and coordination number in $[Co(en)_2Cl_2]^+$.

ANSWER 338

en is neutral and two Cl^- contribute −2. For complex charge +1, x−2=+1, so Co=+3. Two en ligands donate four atoms and two Cl^- donate two: coordination number 6.

QUESTION 339 [H] Write the IUPAC name of $[Cr(H_2O)_4Cl_2]Cl$.

ANSWER 339

Tetraaquadichloridochromium(III) chloride.

QUESTION 340 [M] What is the denticity of ethane-1,2-diamine (en)?

ANSWER 340

Ethane-1,2-diamine (en) is bidentate, denticity 2, donating through its two nitrogen atoms.

QUESTION 341 [H] Predict the geometry and hybridisation of $[Fe(CN)_6]^{3-}$ using the strong- field assumption.

ANSWER 341

Fe in $[Fe(CN)_6]^{3-}$ is Fe^{3+}, d^5. Under strong-field conditions it is low-spin t_{2g}^5 with one unpaired electron; VBT predicts inner-orbital d^2sp^3 octahedral geometry.

QUESTION 342 [M] Why is $[Zn(NH_3)_4]^{2+}$ generally diamagnetic?

ANSWER 342

Zn^{2+} is d^{10}, so all metal d electrons are paired; $[Zn(NH_3)_4]^{2+}$ is therefore diamagnetic in the usual description.

QUESTION 343 [H] Compare the magnetic moments expected for high-spin and low-spin d^6 octahedral complexes.

ANSWER 343

Octahedral d^6 high spin $t_{2g}^4e_g^2$ has four unpaired electrons, μ≈4.90 BM; low spin t_{2g}^6 has none and is diamagnetic (spin-only values).

QUESTION 344 [M] What is the difference between crystal-field splitting energy and pairing energy?

ANSWER 344

Δ is the energy gap between ligand-field orbital sets. Pairing energy P is the energetic cost of placing two electrons in one orbital; comparing Δ and P determines spin state.

QUESTION 345 [H] For an octahedral d^4 ion, compare high-spin and low-spin electron arrangements.

ANSWER 345

Octahedral d^4 high spin: $t_{2g}^3 e_g^1$, four unpaired. Low spin: $t_{2g}^4 e_g^0$, two unpaired; low spin is favoured when $\Delta_o > P$.

QUESTION 346 [M] Give an example of a complex that exhibits optical isomerism.

ANSWER 346

$[Co(en)_3]^{3+}$ is an example; its Δ and Λ forms are non-superimposable mirror images.

QUESTION 347 [H] Explain why cis-$[M(en)_2Cl_2]^+$ can be optically active while its trans form may not be.

ANSWER 347

In cis-$[M(en)_2Cl_2]^+$, the two chelate rings can form a chiral arrangement with Δ/Λ mirror-image forms and no improper symmetry element. The trans arrangement has symmetry that makes it achiral in the usual octahedral case.

QUESTION 348 [M] What is coordination sphere?

ANSWER 348

The coordination sphere is the central metal together with directly attached ligands, conventionally written inside square brackets.

QUESTION 349 [H] Determine the number of ions produced when $[Co(NH_3)_5Cl]Cl_2$ dissolves completely in water, ignoring complex dissociation.

ANSWER 349

$[Co(NH_3)_5Cl]Cl_2 \rightarrow [Co(NH_3)_5Cl]^{2+} + 2Cl^-$, giving three ions per formula unit if the complex ion remains intact.

QUESTION 350 [M] Why do coordination compounds often have specific geometries?

ANSWER 350

Geometry reflects coordination number, ligand size/repulsion, metal electronic structure and ligand-field stabilisation; these favour characteristic arrangements such as tetrahedral, square planar or octahedral.

QUESTION 351 [H] Explain the limitations of VBT in explaining colour and spectra of coordination compounds.

ANSWER 351

VBT gives qualitative hybridisation/geometry and pairing but does not quantitatively explain d-orbital splitting, colours, absorption spectra, spectrochemical ordering or many magnetic trends.

QUESTION 352 [M] What is the chelate effect?

ANSWER 352

The chelate effect is the enhanced thermodynamic stability often observed when a multidentate ligand replaces comparable monodentate ligands, commonly due to favourable entropy and multiple simultaneous bonds.

QUESTION 353 [H] Explain how coordination compounds can mask interfering ions in qualitative analysis.

ANSWER 353

A masking ligand binds an interfering metal ion strongly, lowering its free-ion concentration and preventing its usual precipitation or colour reaction while other ions are tested.

QUESTION 354 [M] Why is $[Ag(NH_3)_2]^+$ important in qualitative analysis?

ANSWER 354

$[Ag(NH_3)_2]^+$ is the soluble diamminesilver(I) complex in Tollens' reagent, used to identify aldehydes by their reduction of Ag(I) to metallic silver.

QUESTION 355 [H] Explain how complex formation can alter the solubility of a sparingly soluble salt.

ANSWER 355

A complexing ligand removes free metal ions from solution. This shifts the dissolution equilibrium of a sparingly soluble salt toward more dissolved ions, increasing apparent solubility.

QUESTION 356 [M] Write the formula of tetraamminecopper(II) sulfate.

ANSWER 356

$[Cu(NH_3)_4]SO_4$.

QUESTION 357 [H] Distinguish between fac and mer isomers of an octahedral $[MA_3B_3]$ complex.

ANSWER 357

In octahedral $[MA_3B_3]$, fac has the three A ligands on one face, all mutually cis. Mer has the three A ligands along a meridian, with one A–A pair trans.

QUESTION 358 [H] For an octahedral d^5 metal ion, compare the number of unpaired electrons in high-spin and low-spin cases.

ANSWER 358

Octahedral d^5 high-spin $t_{2g}^3 e_g^2$ has five unpaired electrons; low-spin $t_{2g}^5 e_g^0$ has one unpaired electron.

UNIT 6: HALOALKANES AND HALOARENES

Questions 359–428 · 70 complete pairs

QUESTION 359 [M] Define haloalkanes and haloarenes, giving one example of each.

ANSWER 359

A haloalkane has halogen bonded to an aliphatic sp^3 carbon (e.g. CH_3CH_2Cl). A haloarene has halogen directly bonded to an aromatic-ring carbon (e.g. chlorobenzene).

QUESTION 360 [M] Write the IUPAC name of $CH_3CHBrCH_2CH_3$.

ANSWER 360

2-Bromobutane.

QUESTION 361 [M] Explain why the carbon–halogen bond is polar.

ANSWER 361

Halogens are more electronegative than carbon, so the C–X bond is polar: $C\delta^+–X\delta^-$.

QUESTION 362 [H] Compare the C–F, C–Cl, C–Br and C–I bond strengths and relate them to reactivity.

ANSWER 362

Bond strength decreases C–F>C–Cl>C–Br>C–I as bond length increases. For substitution, easier C–X cleavage generally gives RI>RBr>RCl≫RF, although mechanism and substrate also matter.

QUESTION 363 [M] What is nucleophilic substitution?

ANSWER 363

Nucleophilic substitution replaces a leaving group (often halide) on carbon with a nucleophile: $R–X+Nu^-→R–Nu+X^-$.

QUESTION 364 [M] Distinguish between SN1 and SN2 reactions.

ANSWER 364

SN1 is two-step, through a carbocation; rate=k[RX], often favoured by tertiary substrate/polar protic solvent and gives racemisation tendency. SN2 is concerted backside attack; rate=k[RX][Nu^-], favoured by less hindered substrate/strong nucleophile and gives inversion.

QUESTION 365 [H] Explain why tertiary haloalkanes generally favour SN1 while primary haloalkanes generally favour SN2 under suitable conditions.

ANSWER 365

Tertiary substrates form relatively stable carbocations and are sterically hindered to backside attack, favouring SN1. Primary substrates form unstable carbocations but allow backside approach, favouring SN2; solvent/nucleophile can alter the outcome.

QUESTION 366 [M] What is racemisation in a nucleophilic substitution reaction?

ANSWER 366

Racemisation is loss of optical purity as an initially optically active substrate forms a mixture of enantiomers, often through attack on both faces of a planar carbocation.

QUESTION 367 [H] Explain the stereochemical outcome of an SN2 reaction at a chiral carbon.

ANSWER 367

SN2 proceeds by backside attack as the C–Nu bond forms and C–X breaks, causing inversion of configuration at the reacting stereogenic centre (Walden inversion).

QUESTION 368 [M] What is optical activity?

ANSWER 368

Optical activity is the ability of a chiral substance to rotate the plane of plane-polarised light; enantiomers rotate it equally in opposite directions.

QUESTION 369 [H] Why can an SN1 reaction at a chiral centre lead to racemisation?

ANSWER 369

SN1 forms a roughly planar carbocation, which can be attacked from either face, producing both configurations. Ion-pair effects often make racemisation incomplete rather than exactly 50:50.

QUESTION 370 [M] Arrange primary, secondary and tertiary haloalkanes in order of SN1 reactivity, with justification.

ANSWER 370

SN1 reactivity generally 3°>2°>1° because increasing alkyl substitution stabilises the rate-determining carbocation.

QUESTION 371 [H] Predict the major product when 2-bromobutane reacts with aqueous KOH and explain the reaction type.

ANSWER 371

Major product is butan-2-ol, $CH_3CH(OH)CH_2CH_3$, by nucleophilic substitution with aqueous OH^- replacing Br^-; secondary-substrate elimination can be a minor competing pathway depending on conditions.

QUESTION 372 [M] What is dehydrohalogenation?

ANSWER 372

Dehydrohalogenation is elimination of HX from adjacent (α and β) carbons of a haloalkane to form an alkene, commonly with alcoholic base and heat.

QUESTION 373 [H] Compare substitution and elimination products when a haloalkane reacts with aqueous KOH versus alcoholic KOH.

ANSWER 373

Aqueous KOH favours nucleophilic substitution to an alcohol. Alcoholic KOH with heat favours β-elimination to an alkene; substrate and conditions can allow competing pathways.

QUESTION 374 [M] Why are haloarenes less reactive than haloalkanes towards nucleophilic substitution?

ANSWER 374

Aryl C–X has partial double-bond character from resonance, the bonded carbon is sp^2, and backside attack/phenyl-cation formation is difficult; haloarenes therefore resist ordinary SN reactions.

QUESTION 375 [H] Explain the partial double-bond character of the C–Cl bond in chlorobenzene using resonance.

ANSWER 375

A chlorine lone pair overlaps with the benzene π system. Resonance contributors place π bonding between C and Cl, giving the aryl C–Cl bond partial double-bond character and shortening/strengthening it.

QUESTION 376 [M] Why does chlorobenzene direct electrophilic substitution to ortho and para positions?

ANSWER 376

Cl donates a lone pair by resonance, increasing electron density mainly at ortho and para positions and stabilising the corresponding arenium intermediates; these positions are therefore directed despite deactivation.

QUESTION 377 [H] Explain why halogens are deactivating but ortho/para-directing in electrophilic aromatic substitution.

ANSWER 377

The strong −I effect withdraws electron density and deactivates the ring. Lone-pair resonance donation stabilises ortho/para substitution intermediates, so halogens remain ortho/para directors.

QUESTION 378 [M] Write the products of nitration of chlorobenzene.

ANSWER 378

Nitration gives ortho- and para-nitrochlorobenzene, with the para isomer generally major because it is less sterically crowded.

QUESTION 379 [M] What is the Wurtz reaction?

ANSWER 379

Wurtz reaction couples two alkyl halides with sodium in dry ether: $2R–X+2Na→R–R+2NaX$.

QUESTION 380 [M] What is the Fittig reaction?

ANSWER 380

Fittig reaction couples aryl halides with sodium in dry ether: $2Ar–X+2Na→Ar–Ar+2NaX$.

QUESTION 381 [M] What is the Wurtz–Fittig reaction?

ANSWER 381

Wurtz–Fittig reaction couples an aryl and an alkyl halide with sodium in dry ether: $Ar–X+R–X+2Na→Ar–R+2NaX$.

QUESTION 382 [H] Predict the products when bromobenzene reacts with sodium in dry ether in the presence of methyl bromide.

ANSWER 382

Principal Wurtz–Fittig product is toluene: $C_6H_5Br + CH_3Br + 2Na \rightarrow C_6H_5CH_3 + 2NaBr$. Homocoupling side-products (biphenyl and ethane) may also form.

QUESTION 383 [M] How can a haloalkane be converted into an alcohol?

ANSWER 383

Heat the haloalkane with aqueous NaOH or KOH to replace X by OH: $R\text{–}X + OH^- \rightarrow R\text{–}OH + X^-$.

QUESTION 384 [H] Suggest a route to convert ethanol into bromoethane and state a suitable reagent.

ANSWER 384

Convert ethanol to bromoethane with HBr (often generated from NaBr/KBr and concentrated H_2SO_4) or use PBr_3: $CH_3CH_2OH \rightarrow CH_3CH_2Br$.

QUESTION 385 [M] What is the difference between allylic and vinylic halides?

ANSWER 385

An allylic halide has X on an sp^3 carbon adjacent to C=C; a vinylic halide has X directly attached to an alkene (sp^2) carbon.

QUESTION 386 [H] Explain why vinyl chloride is less reactive than ethyl chloride towards nucleophilic substitution.

ANSWER 386

Vinyl C–Cl has partial double-bond character through conjugation, is shorter/stronger at an sp^2 carbon and resists backside attack; ethyl chloride has an ordinary sp^3 C–Cl bond.

QUESTION 387 [M] State one use and one environmental concern associated with dichloromethane.

ANSWER 387 · NEEDS QUALIFICATION

Dichloromethane is used as an industrial solvent and in paint strippers. Emissions contribute to stratospheric ozone depletion despite its relatively short atmospheric lifetime (Australian Industrial Chemicals Introduction Scheme evaluation, 2021).

QUESTION 388 [M] State one use and one health concern associated with chloroform.

ANSWER 388

Chloroform has been used as a solvent and historically as an anaesthetic. It can damage liver/kidneys and depress the central nervous system; it is unsafe for unsupervised exposure.

QUESTION 389 [M] Why is carbon tetrachloride environmentally problematic?

ANSWER 389

CCl_4 is toxic and persistent, and chlorine radicals released in the stratosphere catalytically deplete ozone; its production/use has been heavily restricted.

QUESTION 390 [M] What is iodoform and what is one historical use of it?

ANSWER 390

Iodoform is triiodomethane, CHI_3, a yellow crystalline solid historically used as an antiseptic/dressing because it releases iodine-containing species.

QUESTION 391 [M] What are freons and why have many been phased out?

ANSWER 391

Freons are chlorofluorocarbon (CFC) refrigerants/propellants. Stable CFCs reach the stratosphere and release chlorine radicals that deplete ozone; many were phased out under the Montreal Protocol.

QUESTION 392 [M] Why is DDT considered an environmental pollutant?

ANSWER 392

DDT is persistent, fat-soluble and bioaccumulates/biomagnifies, harming wildlife (notably birds) and potentially exposing people through food chains.

QUESTION 393 [H] Compare the environmental persistence and bioaccumulation concerns of DDT and many modern pesticides.

ANSWER 393 · NEEDS QUALIFICATION

DDT is notably persistent and bioaccumulative. Many newer pesticides are designed to degrade faster, but persistence and bioaccumulation vary by compound; some modern pesticides remain hazardous, so a blanket claim that all are safer is not justified.

QUESTION 394 [M] How does the nature of the halogen affect the boiling point of haloalkanes?

ANSWER 394

For the same alkyl group, molar mass and polarizability generally rise F→Cl→Br→I, strengthening dispersion forces; boiling points usually rise in that order (with structural exceptions).

QUESTION 395 [H] Explain why branching generally lowers the boiling point of isomeric haloalkanes.

ANSWER 395

Branching makes isomers more compact, reducing intermolecular contact surface and London dispersion forces; less energy is required to separate molecules, so boiling point generally falls.

QUESTION 396 [M] Why are haloalkanes only slightly soluble in water?

ANSWER 396

Haloalkanes cannot hydrogen-bond effectively with water, and their hydrophobic alkyl group makes hydration energetically unfavourable; only slight amounts dissolve.

QUESTION 397 [H] Explain why haloalkanes can be soluble in organic solvents despite having a polar C–X bond.

ANSWER 397

Many organic solvents have compatible polarizability and dispersion interactions with haloalkanes, so solute–solvent attractions can offset disruption of solvent–solvent interactions despite C–X bond polarity.

QUESTION 398 [M] What is a Grignard reagent?

ANSWER 398

A Grignard reagent is an organomagnesium halide, RMgX, containing a strongly polar C–Mg bond and behaving as a powerful nucleophilic/basic carbon reagent.

QUESTION 399 [H] Why must Grignard reagents be prepared and used under anhydrous conditions?

ANSWER 399

Water protonates the carbon end and destroys the reagent: $RMgX + H_2O \rightarrow RH + Mg(OH)X$. Dry ether coordinates/stabilises Mg and must also be anhydrous.

QUESTION 400 [M] Write the reaction of bromoethane with magnesium in dry ether.

ANSWER 400

$CH_3CH_2Br + Mg \rightarrow CH_3CH_2MgBr$ in dry ether.

QUESTION 401 [H] Predict the product when a Grignard reagent reacts with carbon dioxide followed by acidic hydrolysis.

ANSWER 401

$RMgX + CO_2 \rightarrow RCO_2MgX$; acidic hydrolysis gives RCOOH. The carboxylic acid has one more carbon than R.

QUESTION 402 [M] How can the presence of halogen in an organic compound be tested after suitable preparation of the sample?

ANSWER 402

Use sodium fusion (Lassaigne's test) to convert halogen to NaX in the extract; acidify with dilute HNO_3, then add $AgNO_3$. AgCl is white, AgBr pale yellow and AgI yellow; confirmatory solubility tests can distinguish them.

QUESTION 403 [H] Explain why direct treatment of an organic halide with aqueous silver nitrate may not give a reliable result in every case.

ANSWER 403 · NEEDS QUALIFICATION

Covalently bound halogen may not release X^- simply on contact with aqueous $AgNO_3$, especially for aryl/vinyl halides. The test depends on hydrolysis/ionisation and solubility, so sample preparation and suitable reaction conditions are needed.

QUESTION 404 [M] Distinguish between nucleophiles and bases.

ANSWER 404

A nucleophile donates an electron pair to an electrophilic atom (often carbon) to form a bond. A base accepts a proton; many species can do both, but the attacked site differs.

QUESTION 405 [H] Explain how solvent polarity and solvent proticity can affect SN1 and SN2 reactions.

ANSWER 405

Polar protic solvents stabilise ions/carbocations and solvate anionic nucleophiles, often favouring SN1 and slowing SN2. Polar aprotic solvents solvate cations but leave nucleophiles reactive, often accelerating SN2.

QUESTION 406 [M] What is the role of a leaving group in nucleophilic substitution?

ANSWER 406

A leaving group departs from carbon with the bonding electron pair, enabling substitution. Stable, weakly basic anions generally leave more readily.

QUESTION 407 [H] Arrange typical halide leaving groups in order of leaving-group ability and justify the trend.

ANSWER 407

Typical ability: $I^- > Br^- > Cl^- \gg F^-$. Larger, more polarizable anions are more stable and weaker bases, and the C–I bond is weakest; the exact rate still depends on substrate and mechanism.

QUESTION 408 [M] Write the IUPAC name of $CH_3CH_2CHClCH_3$.

ANSWER 408

2-Chlorobutane.

QUESTION 409 [H] Predict the major product of 2-bromopropane with alcoholic KOH and explain the mechanism.

ANSWER 409

Propene is the major product: $CH_3CHBrCH_3 + KOH(alc) \rightarrow CH_3CH=CH_2 + KBr + H_2O$. Alcoholic base and heat favour concerted E2 β-elimination.

QUESTION 410 [M] Why does tertiary butyl chloride react rapidly by SN1 in a polar protic solvent?

ANSWER 410

Tert-butyl chloride forms a relatively stable tertiary carbocation; a polar protic solvent stabilises the developing ions, accelerating the SN1 ionisation step.

QUESTION 411 [H] Explain the energy profile of SN1 and SN2 mechanisms.

ANSWER 411

SN1 has two transition states separated by a carbocation intermediate; the first, ionisation barrier is often rate-determining. SN2 has one concerted transition state and one barrier, with simultaneous C–Nu formation and C–X cleavage.

QUESTION 412 [M] What is inversion of configuration?

ANSWER 412

Inversion of configuration is reversal of the spatial arrangement at a stereogenic carbon, as in backside SN2 attack (Walden inversion).

QUESTION 413 [H] A chiral haloalkane undergoes substitution with a strong nucleophile in a polar aprotic solvent. Predict the likely stereochemical outcome.

ANSWER 413

Likely SN2: the strong nucleophile attacks from the backside in polar aprotic solvent, producing inversion at the reacting chiral centre, provided steric hindrance does not prevent SN2.

QUESTION 414 [M] Why is chlorobenzene resistant to hydrolysis under ordinary conditions?

ANSWER 414

Chlorobenzene's aryl C–Cl bond has resonance partial double-bond character and is attached to an sp^2 carbon; ordinary backside substitution is difficult, so hydrolysis requires much harsher conditions than for alkyl chlorides.

QUESTION 415 [H] Explain how electron-withdrawing groups at ortho and para positions facilitate nucleophilic substitution in haloarenes.

ANSWER 415

Ortho/para electron-withdrawing groups stabilise the negatively charged Meisenheimer addition intermediate by resonance delocalisation, lowering the barrier for nucleophilic aromatic substitution.

QUESTION 416 [M] What is the product of hydrolysis of bromoethane with aqueous KOH?

ANSWER 416

Bromoethane hydrolyses to ethanol: $CH_3CH_2Br + OH^- \rightarrow CH_3CH_2OH + Br^-$.

QUESTION 417 [H] Give a sequence to convert propene into 1-bromopropane, noting the required conditions.

ANSWER 417

Add HBr to propene in the presence of peroxide (ROOR), usually with heat/light; radical anti-Markovnikov addition gives $CH_3CH_2CH_2Br$ (1-bromopropane).

QUESTION 418 [M] What is the role of dry ether in the Wurtz reaction?

ANSWER 418

Dry ether solvates/stabilises the sodium and organometallic intermediates and provides an inert medium; moisture would consume sodium or protonate reactive species.

QUESTION 419 [H] Why can Wurtz reactions of two different haloalkanes give a mixture of alkanes?

ANSWER 419

Each haloalkane can couple with either partner, producing both cross-coupled and homocoupled products. The mixture is difficult to avoid unless the reactants are chosen carefully.

QUESTION 420 [M] Why are CFCs harmful to stratospheric ozone?

ANSWER 420

UV light breaks C–Cl bonds in stratospheric CFCs, releasing chlorine radicals. These radicals catalytically convert ozone to oxygen and are regenerated.

QUESTION 421 [H] Explain the chain mechanism by which chlorine radicals destroy ozone, at a conceptual level.

ANSWER 421

$Cl\cdot + O_3 \rightarrow ClO\cdot + O_2$; then $ClO\cdot + O\cdot \rightarrow Cl\cdot + O_2$. Chlorine radical is regenerated, so the net reaction is $O_3 + O \rightarrow 2O_2$ and one radical can destroy many ozone molecules.

QUESTION 422 [M] What is the difference between a geminal and a vicinal dihalide?

ANSWER 422

A geminal dihalide has both halogen atoms on the same carbon; a vicinal dihalide has them on adjacent carbons.

QUESTION 423 [H] Compare the reactivity of allyl chloride, ethyl chloride and vinyl chloride toward nucleophilic substitution.

ANSWER 423

Allyl chloride is highly reactive because its transition state/carbocation is resonance-stabilised; ethyl chloride has ordinary primary-halide reactivity; vinyl chloride is much less reactive because its sp^2 C–Cl bond has partial double-bond character. Typical order: allyl chloride > ethyl chloride ≫ vinyl chloride.

QUESTION 424 [M] Why are organohalogen compounds often denser than corresponding hydrocarbons?

ANSWER 424

Halogens are much heavier and more polarizable than hydrogen, so replacing H with one or more halogens raises molar mass and often density relative to the corresponding hydrocarbon.

QUESTION 425 [H] Explain how a chiral centre can be formed during a substitution reaction and how its configuration is assigned.

ANSWER 425 · NEEDS QUALIFICATION

A new stereogenic centre forms if substitution gives a carbon attached to four different groups. Assign R/S by CIP priorities (atomic number, then successive ties); determine whether the mechanism gives inversion, racemisation or another stereochemical outcome separately.

QUESTION 426 [M] What is the major organic product when bromoethane reacts with potassium cyanide?

ANSWER 426

Propanenitrile (ethyl cyanide), $CH_3CH_2C{\equiv}N$; KCN attacks through carbon, adding one carbon to the chain.

QUESTION 427 [H] Compare the products obtained when bromoethane reacts with KCN and AgCN.

ANSWER 427

KCN is ionic and supplies free CN^-, which attacks through carbon to give propanenitrile, CH_3CH_2CN. Covalent AgCN favours nitrogen-end attack, giving ethyl isocyanide, CH_3CH_2NC.

QUESTION 428 [H] Explain why the rate of an SN2 reaction depends on both haloalkane and nucleophile concentrations.

ANSWER 428

SN2 is a single concerted collision of haloalkane and nucleophile, so its rate law is rate=$k[RX][Nu^-]$; increasing either reactant concentration increases the frequency of productive encounters.

UNIT 7: ALCOHOLS, PHENOLS AND ETHERS

Questions 429–500 · 72 complete pairs

QUESTION 429 [M] Define alcohols, phenols and ethers with one example of each.

ANSWER 429

Alcohols have –OH on an aliphatic carbon (ethanol, CH_3CH_2OH); phenols have –OH directly on an aromatic ring (phenol, C_6H_5OH); ethers have oxygen between carbon groups (diethyl ether, $C_2H_5OC_2H_5$).

QUESTION 430 [M] Write the IUPAC name of $CH_3CH_2CH(OH)CH_3$.

ANSWER 430

Butan-2-ol.

QUESTION 431 [M] Distinguish between primary, secondary and tertiary alcohols.

ANSWER 431

In a primary alcohol the carbon bearing OH is attached to one other carbon; in a secondary alcohol to two; in a tertiary alcohol to three.

QUESTION 432 [M] Why do alcohols have higher boiling points than comparable alkanes?

ANSWER 432

Alcohol molecules form intermolecular hydrogen bonds through –OH groups. More energy is needed to separate them than comparable alkanes, which rely mainly on dispersion forces.

QUESTION 433 [H] Explain how hydrogen bonding affects the boiling points and water solubilities of alcohols.

ANSWER 433

Intermolecular hydrogen bonds raise alcohol boiling points. The –OH group hydrogen-bonds with water, making lower alcohols soluble; as the hydrophobic alkyl chain grows, water solubility decreases.

QUESTION 434 [M] Describe one method for preparing ethanol from ethene.

ANSWER 434

Hydrate ethene with steam over a phosphoric-acid catalyst at elevated temperature/pressure: $CH_2{=}CH_2 + H_2O \rightleftharpoons CH_3CH_2OH$.

QUESTION 435 [M] How can a haloalkane be converted into an alcohol?

ANSWER 435

Heat a haloalkane with aqueous NaOH or KOH; OH^- substitutes for halide to give the alcohol, $R{-}X + OH^- \rightarrow R{-}OH + X^-$.

QUESTION 436 [H] Explain the preparation of alcohols by reduction of aldehydes and ketones.

ANSWER 436

Aldehydes reduce to primary alcohols and ketones to secondary alcohols, e.g. with $NaBH_4$ or $LiAlH_4$ followed by work-up: $RCHO \rightarrow RCH_2OH$; $R_2CO \rightarrow R_2CHOH$.

QUESTION 437 [M] What is the oxidation product of a primary alcohol under controlled oxidation?

ANSWER 437

A primary alcohol under controlled oxidation gives an aldehyde, $RCH_2OH + [O] \rightarrow RCHO + H_2O$; distil promptly or use a mild reagent to limit further oxidation to the carboxylic acid.

QUESTION 438 [H] Compare the oxidation products of primary, secondary and tertiary alcohols under common conditions.

ANSWER 438

Primary alcohols oxidise first to aldehydes and, under stronger/continued conditions, carboxylic acids. Secondary alcohols give ketones. Tertiary alcohols resist mild oxidation; strong conditions may cleave C–C bonds.

QUESTION 439 [M] What is the Lucas test? What does it help distinguish?

ANSWER 439

Lucas test uses concentrated HCl with anhydrous $ZnCl_2$ to convert alcohols to insoluble alkyl chlorides. Turbidity time distinguishes primary, secondary and tertiary alcohols approximately.

QUESTION 440 [H] Explain the relative rates of reaction of primary, secondary and tertiary alcohols with Lucas reagent.

ANSWER 440

Tertiary alcohols turn cloudy almost immediately, secondary alcohols after minutes, while primary alcohols show little/no turbidity at room temperature; carbocation stability controls the usual SN1-like rate order 3°>2°>1°.

QUESTION 441 [M] What happens when ethanol reacts with sodium metal?

ANSWER 441

Ethanol forms sodium ethoxide and hydrogen: $2C_2H_5OH + 2Na \rightarrow 2C_2H_5ONa + H_2\uparrow$.

QUESTION 442 [M] What is dehydration of an alcohol?

ANSWER 442

Dehydration removes H_2O from an alcohol, usually forming an alkene under acid/heat or an ether under suitable lower-temperature conditions.

QUESTION 443 [H] Describe the mechanism of acid-catalysed dehydration of ethanol to ethene.

ANSWER 443 · NEEDS QUALIFICATION

NCERT's Class XII scheme for ethanol dehydration at about 443 K is: protonate –OH; lose H_2O (shown as the slow step forming a formal ethyl-cation intermediate); then remove a β-H to form $CH_2=CH_2$ and regenerate acid. Qualification: this is the textbook mechanism to report for the syllabus, but the drawn primary ethyl cation should not be interpreted as a stable, freely solvated intermediate; the actual elimination is strongly coupled.

QUESTION 444 [M] Why is concentrated sulfuric acid used in the dehydration of alcohols?

ANSWER 444

Concentrated H_2SO_4 protonates –OH so it can leave as water and, as a dehydrating agent, favours water removal; temperature helps control alkene versus ether formation.

QUESTION 445 [H] Explain why tertiary alcohols generally dehydrate more readily than primary alcohols.

ANSWER 445

Tertiary alcohols can form more stable carbocations after protonated water leaves, facilitating E1 dehydration; secondary are intermediate and primary are least favourable (often eliminate by a concerted route).

QUESTION 446 [M] What is the product when ethanol is heated with excess concentrated sulfuric acid at about 413 K?

ANSWER 446 · NEEDS QUALIFICATION

Under the standard NCERT conditions, ether formation predominates near 413 K: $2C_2H_5OH \rightarrow C_2H_5OC_2H_5 + H_2O$ (ethoxyethane). Qualification: the prompt says "excess concentrated H_2SO_4" rather than specifying the ethanol-rich conditions normally used for ether formation, so the product claim is conditional on the usual low-temperature dehydration setup.

QUESTION 447 [H] Compare the products formed by heating ethanol with concentrated sulfuric acid at about 413 K and 443 K.

ANSWER 447

At about 413 K, intermolecular dehydration gives diethyl ether. At about 443 K with concentrated H_2SO_4, intramolecular elimination gives ethene, $CH_2=CH_2$.

QUESTION 448 [M] Define phenol and write its formula.

ANSWER 448

Phenol is hydroxybenzene, with an –OH group directly attached to a benzene ring; formula C_6H_5OH (C_6H_6O).

QUESTION 449 [M] Why is phenol more acidic than ethanol?

ANSWER 449

Phenol loses H^+ to form phenoxide, whose negative charge is delocalised into the aromatic ring. Ethoxide has no comparable resonance stabilisation, so ethanol is much less acidic.

QUESTION 450 [H] Explain the greater stability of the phenoxide ion compared with the ethoxide ion.

ANSWER 450

Phenoxide's negative charge is delocalised over oxygen and ortho/para ring positions by resonance. Ethoxide's charge remains localised mainly on oxygen, so phenoxide is more stable.

QUESTION 451 [M] How does an electron-withdrawing group affect the acidity of phenol?

ANSWER 451

An electron-withdrawing group stabilises phenoxide by inductive and/or resonance effects, especially at ortho/para positions, and increases phenol acidity.

QUESTION 452 [H] Compare the acidity of phenol, p-nitrophenol and p-cresol with justification.

ANSWER 452

Acidity: p-nitrophenol>phenol>p-cresol. The para nitro group withdraws and stabilises phenoxide; the methyl group donates electron density and destabilises it.

QUESTION 453 [M] Describe the preparation of phenol from chlorobenzene.

ANSWER 453 · CORRECTED

Dow process: $C_6H_5Cl + 2NaOH(aq) \rightarrow C_6H_5ONa + NaCl + H_2O$ (623 K, ~300 atm); then acidify C_6H_5ONa with H^+/dilute acid to obtain C_6H_5OH.

QUESTION 454 [M] What happens when phenol reacts with bromine water?

ANSWER 454

Bromine water rapidly substitutes at the activated ortho and para positions to give 2,4,6-tribromophenol as a white precipitate, decolourising bromine water.

QUESTION 455 [H] Explain why phenol undergoes electrophilic substitution more readily than benzene.

ANSWER 455

The –OH group donates a lone pair into the aromatic ring by resonance, increasing ring electron density and stabilising ortho/para arenium-ion intermediates; phenol reacts faster than benzene.

QUESTION 456 [M] What is the Kolbe–Schmitt reaction?

ANSWER 456

Kolbe–Schmitt reaction: sodium phenoxide reacts with CO_2 under pressure/heat to form sodium salicylate; acidification yields salicylic acid (2-hydroxybenzoic acid).

QUESTION 457 [M] What is the Reimer–Tiemann reaction?

ANSWER 457

Reimer–Tiemann reaction formylates phenol mainly at the ortho position using $CHCl_3$ and aqueous NaOH, followed by acid work-up, to give salicylaldehyde.

QUESTION 458 [H] Write the main product of the Reimer–Tiemann reaction of phenol and explain the role of the reagent.

ANSWER 458

Main product is 2-hydroxybenzaldehyde (salicylaldehyde). $CHCl_3$/NaOH generates dichlorocarbene, $:CCl_2$, which introduces the formyl group after hydrolysis.

QUESTION 459 [M] What happens when phenol reacts with dilute nitric acid?

ANSWER 459

Dilute HNO_3 gives a mixture of 2-nitrophenol and 4-nitrophenol by electrophilic substitution; the para isomer is often major.

QUESTION 460 [H] Compare the products of nitration of phenol under dilute and concentrated nitric acid conditions.

ANSWER 460

Dilute HNO_3 gives mainly ortho- and para-nitrophenol. Concentrated HNO_3 causes further nitration to 2,4,6-trinitrophenol (picric acid).

QUESTION 461 [M] How can phenol be distinguished from ethanol using a simple chemical test?

ANSWER 461

Add neutral $FeCl_3$: phenol gives a violet/purple coloured complex, whereas ethanol does not give this characteristic colour.

QUESTION 462 [M] Define ethers and give the general structure of an ether.

ANSWER 462

An ether has an oxygen atom bonded to two carbon groups: $R–O–R'$ (which may be identical or different).

QUESTION 463 [M] Write the IUPAC name of $CH_3OCH_2CH_3$.

ANSWER 463

Methoxyethane.

QUESTION 464 [M] Describe the Williamson ether synthesis.

ANSWER 464

Williamson synthesis: react a sodium alkoxide/phenoxide with a suitable primary alkyl halide by SN2: $RONa + R'X \rightarrow ROR' + NaX$.

QUESTION 465 [H] Why is a primary alkyl halide generally preferred in Williamson ether synthesis?

ANSWER 465

A primary halide is least sterically hindered and undergoes SN2 substitution readily. Secondary/tertiary halides increasingly favour E2 elimination with the alkoxide base.

QUESTION 466 [H] Predict the major product when sodium ethoxide reacts with 2- bromopropane and explain the competing reaction.

ANSWER 466

Propene is usually the major product by E2 elimination because 2-bromopropane is secondary and ethoxide is a strong base. A smaller amount of ethyl isopropyl ether may form by competing SN2 substitution.

QUESTION 467 [M] What happens when an ether is heated with excess HI?

ANSWER 467

Excess hot HI cleaves both C–O bonds of a dialkyl ether: $C_2H_5OC_2H_5 + 2HI \rightarrow 2C_2H_5I + H_2O$.

QUESTION 468 [H] Explain the cleavage of an unsymmetrical ether by HI and how the products depend on the alkyl groups.

ANSWER 468

Protonate ether oxygen, then I^- usually attacks the less hindered alkyl group by SN2. If one side is tertiary, cleavage at that side can proceed by SN1; in aryl–alkyl ethers the alkyl–O bond breaks, giving phenol and alkyl iodide.

QUESTION 469 [M] Why do ethers have lower boiling points than isomeric alcohols?

ANSWER 469

Alcohols can donate and accept intermolecular hydrogen bonds; ethers can accept but cannot donate O–H hydrogen bonds. Weaker intermolecular association gives ethers lower boiling points than isomeric alcohols.

QUESTION 470 [M] State one use of diethyl ether.

ANSWER 470

Diethyl ether has been used as a solvent and historically as a general anaesthetic; its flammability and peroxide hazard require careful handling.

QUESTION 471 [H] Explain why ethers can form explosive peroxides on prolonged exposure to air and light.

ANSWER 471

Oxygen and light can initiate radical autoxidation at α-C–H positions of ethers, forming explosive hydroperoxides/peroxides that may concentrate during storage or distillation.

QUESTION 472 [M] What is the effect of methanol on human health?

ANSWER 472

Methanol is metabolised to formaldehyde and formate, causing metabolic acidosis, optic-nerve damage/blindness and potentially death; even small ingestions are medical emergencies.

QUESTION 473 [M] Why is ethanol used as a solvent in many products?

ANSWER 473

Ethanol dissolves many polar and moderately non-polar substances, mixes with water, evaporates readily and is relatively accessible, making it useful in formulations, cleaning and extraction.

QUESTION 474 [H] Compare methanol and ethanol in terms of preparation, uses and toxicity.

ANSWER 474

Methanol is made mainly from synthesis gas and used as a feedstock/solvent/fuel; it is highly toxic and can cause blindness or death. Ethanol is made by fermentation or ethene hydration and used as solvent/fuel; drinking it is intoxicating and excessive exposure is harmful.

QUESTION 475 [M] What is esterification of an alcohol?

ANSWER 475

Esterification is formation of an ester from an alcohol and a carboxylic acid, usually reversibly under acid catalysis: $RCOOH + R'OH \rightleftharpoons RCOOR' + H_2O$.

QUESTION 476 [H] Write the reaction of ethanol with ethanoic acid and state the role of the acid catalyst.

ANSWER 476

$CH_3COOH + C_2H_5OH \rightleftharpoons CH_3COOC_2H_5 + H_2O$. Acid catalyses carbonyl activation and proton transfers; it accelerates attainment of equilibrium but does not change the equilibrium constant.

QUESTION 477 [M] How does the solubility of alcohols in water change with increasing carbon-chain length?

ANSWER 477

Water solubility generally decreases as the carbon chain length increases: the non-polar hydrophobic portion becomes larger relative to the hydrogen-bonding –OH group.

QUESTION 478 [H] Explain why phenol is only moderately soluble in water despite its ability to form hydrogen bonds.

ANSWER 478

Phenol's –OH can hydrogen-bond with water, but its large hydrophobic phenyl ring is poorly hydrated; this limits solubility compared with lower alcohols.

QUESTION 479 [M] How can primary, secondary and tertiary alcohols be distinguished by oxidation behaviour?

ANSWER 479

On oxidation, primary alcohols give aldehydes and then carboxylic acids; secondary alcohols give ketones; tertiary alcohols resist ordinary mild oxidation and require harsh conditions that may cleave C–C bonds.

QUESTION 480 [M] Write the IUPAC name of $(CH_3)_3COH$.

ANSWER 480

2-Methylpropan-2-ol.

QUESTION 481 [H] Predict the major product when 2-methylpropan-2-ol is dehydrated under acidic conditions.

ANSWER 481

2-Methylpropene, $(CH_3)_2C{=}CH_2$, is the major alkene from acid-catalysed dehydration of 2-methylpropan-2-ol.

QUESTION 482 [M] Why does phenol react with sodium hydroxide but ethanol does not react appreciably?

ANSWER 482

Phenol ($pK_a \approx 10$) is acidic enough for OH^- to deprotonate it, forming resonance-stabilised phenoxide and water. Ethanol ($pK_a \approx 16$) is much weaker; OH^- does not appreciably deprotonate it in water.

QUESTION 483 [H] Explain how resonance and inductive effects together influence the acidity of substituted phenols.

ANSWER 483

Electron-withdrawing −I groups stabilise phenoxide and increase acidity; at ortho/para positions, −M groups can further delocalise charge by resonance. Electron-donating +I/+M groups destabilise phenoxide and reduce acidity; effects depend on position.

QUESTION 484 [M] What is the product when phenol reacts with bromine in a non-polar solvent under controlled conditions?

ANSWER 484

Controlled bromination in a non-polar solvent (e.g. Br_2 in CS_2, low temperature) gives mainly p-bromophenol, with some o-bromophenol; conditions limit further substitution.

QUESTION 485 [H] Explain the difference between bromination of phenol in bromine water and in a non-polar solvent.

ANSWER 485

Bromine water gives 2,4,6-tribromophenol rapidly because phenoxide-like resonance strongly activates all ortho/para positions. In a non-polar solvent under controlled conditions, monobromination occurs, with para usually favoured over ortho.

QUESTION 486 [M] How is anisole prepared by Williamson ether synthesis?

ANSWER 486

React sodium phenoxide with a primary methyl halide, e.g. $C_6H_5ONa + CH_3I \rightarrow C_6H_5OCH_3 + NaI$ (anisole), by SN2 at methyl.

QUESTION 487 [H] Predict the products of anisole cleavage with excess HI and explain which bond is cleaved first.

ANSWER 487

Anisole + HI gives phenol and CH_3I: I^- attacks the methyl group by SN2 after oxygen protonation. The aryl $C(sp^2)$–O bond is not cleaved by ordinary SN2, even with excess HI.

QUESTION 488 [M] Why is the C–O bond in phenol shorter than a typical C–O single bond in an alcohol?

ANSWER 488

A phenol oxygen lone pair is conjugated with the aromatic π system, giving the aryl C–O bond partial double-bond character; this makes it shorter than a typical alcohol C–O single bond.

QUESTION 489 [H] Explain why alcohol dehydration generally follows the order tertiary > secondary > primary under acidic conditions.

ANSWER 489

Acid protonates OH, allowing water to leave. Tertiary carbocations are most stable, then secondary, then primary, so ionisation/E1 dehydration is generally 3°>2°>1°; primary alcohols often use concerted elimination instead.

QUESTION 490 [M] What is the product of oxidation of propan-2-ol?

ANSWER 490

Propan-2-ol oxidises to propanone (acetone), CH_3COCH_3.

QUESTION 491 [H] How can 1-propanol be distinguished from 2-propanol by oxidation products?

ANSWER 491

Oxidise under controlled conditions: 1-propanol gives propanal (and with stronger/continued oxidation, propanoic acid); 2-propanol gives propanone. The aldehyde/acid versus ketone products distinguish them.

QUESTION 492 [M] What is the role of acid in esterification?

ANSWER 492

Acid protonates/activates the carbonyl and facilitates nucleophilic addition and proton transfers, speeding esterification. It is regenerated and does not alter the equilibrium position.

QUESTION 493 [H] Give a reaction sequence to convert phenol into salicylic acid.

ANSWER 493

Treat phenol with NaOH to form sodium phenoxide; react with CO_2 under pressure/heat to form sodium salicylate; acidify to obtain salicylic acid. This is the Kolbe–Schmitt reaction.

QUESTION 494 [M] What is the product of phenol with neutral ferric chloride solution?

ANSWER 494

Phenol gives a violet/purple coloured ferric–phenoxide complex with neutral $FeCl_3$.

QUESTION 495 [H] Explain the limitations of using the ferric chloride test as the only evidence for phenol.

ANSWER 495

The colour is not uniquely conclusive: enols and some other compounds can also complex Fe^{3+}, while pH, concentration, oxidation and sample colour affect observations. Confirm with an independent test or spectroscopy.

QUESTION 496 [M] Why are lower alcohols miscible with water?

ANSWER 496

Lower alcohols have a relatively small hydrophobic alkyl group and an –OH group that hydrogen-bonds strongly with water, making mixing favourable and often miscible.

QUESTION 497 [H] Explain the effect of branching on the boiling points of isomeric alcohols.

ANSWER 497

Branching makes alcohol isomers more compact, reducing contact area and dispersion forces; boiling points generally fall with branching. Hydrogen bonding remains, so exact trends can show exceptions.

QUESTION 498 [M] What is the major product when diethyl ether reacts with excess HI on heating?

ANSWER 498

Excess hot HI cleaves both ether bonds: $(C_2H_5)_2O + 2HI \rightarrow 2C_2H_5I + H_2O$; ethyl iodide is the major organic product.

QUESTION 499 [H] Explain why ethers are relatively unreactive toward bases but can be cleaved by strong hydrogen halides.

ANSWER 499

Ethers have no acidic proton, so ordinary bases do not readily break their C–O bonds. Strong HI/HBr protonates oxygen; the resulting oxonium ion is attacked by halide, cleaving a C–O bond.

QUESTION 500 [H] Predict the products when phenol reacts with aqueous NaOH and then with CO_2 under pressure followed by acidification. Name the reaction.

ANSWER 500

NaOH deprotonates phenol to sodium phenoxide, C_6H_5ONa. CO_2 under pressure followed by acidification gives salicylic acid (2-hydroxybenzoic acid); this is the Kolbe–Schmitt reaction.